

Context-Dependent Benefit–Loss Games: Class Robustness and Equilibrium Persistence

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Abstract

We study finite two-player games in which a common strategic form is evaluated by context-specific actor/affected-side payoff maps. The signs of these payoffs induce four relational classes—intelligent, hapless, malicious, and destructive—that apply to profiles rather than persons. We characterize class-preserving translations, define normalized Hamming sensitivity between contextual class maps, and derive exact robustness radii for class and pure-equilibrium sets. The main strategic result concerns an intelligent and a malicious pure Nash equilibrium in a 2×2 game. They cannot share the actor’s action. If they share the affected side’s action, actor indifference is forced, the intelligent equilibrium Pareto- and welfare-dominates the malicious equilibrium, and coexistence is a knife-edge. Robust coexistence is possible only in the diagonal configuration under strict equilibrium and welfare inequalities; an explicit game has certified open-ball radius $3/4$. Three synthetic benchmarks illustrate how class robustness differs from Nash and dynamic robustness under an exponential-replicator update. Reproducible audits enumerate all $5^8 = 390,625$ games on a five-value payoff grid and apply seeded continuous perturbations. The exhaustive audit finds no counterexample, while the perturbations preserve the business class map locally but destroy its original two-equilibrium set almost surely. The examples are formal benchmarks, not empirical estimates.

Keywords: contextual valuation; game theory; Nash equilibrium; robustness; welfare; population dynamics; social modeling.

1 Introduction

Social actions are rarely evaluated in a vacuum. A public correction, a disciplinary warning, a strategic sacrifice, or a managerial order may have different meanings in a company, a classroom, or a sport team. The same abstract action may be perceived as efficient leadership in one context, humiliation in another, and tactical discipline in a third. This observation motivates the present paper: if an interaction is evaluated through gains and losses, then its classification should depend not only on its formal strategic shape, but also on the context in which the payoffs are assigned.

Cipolla’s benefit–loss plane provides a compact representation of this idea [3]. An action performed by an agent can be represented by a point $(x, y) \in \mathbb{R}^2$, where x is the gain or loss for the acting agent and y is the gain or loss for the affected agent or group. The four main sign regions are interpreted as intelligent $(+, +)$, hapless $(-, +)$, malicious $(+, -)$, and stupid or destructive $(-, -)$ interactions. In the formal model developed here we use the term *destructive*, not as a psychological diagnosis, but as a relational class of interactions producing non-compensating harm. The object of study is not a person; it is a strategic profile under a contextual valuation.

Existing formalizations inspired by Cipolla have explored evolutionary games, imitation dynamics, spatial interaction, agent-based modeling, and biophysical interpretations [12, 2, 25, 18]. These works show that the benefit–loss plane can generate nontrivial dynamics once embedded into population games. However, a complementary structural question remains underdeveloped:

What does it mean to transport the same strategic form from one social context to another while preserving, or failing to preserve, the benefit–loss class, pure-equilibrium membership, and population-level consequences of its interactions?

This paper addresses that question by introducing *context-dependent benefit–loss games*. A contextual benefit–loss game consists of a finite strategic form, a context, and a valuation map assigning to each strategic profile a point in the benefit–loss plane. Different contexts may share the same strategic form while assigning different payoff vectors to the same profiles. Thus, context is represented not by changing the abstract strategies themselves, but by changing the valuation map that interprets those strategies.

The paper is organized around a single mathematical object, the contextual valuation map $V_C : A \times B \rightarrow \mathbb{R}^2$. The contribution lies in showing that this map simultaneously induces three connected structures: a benefit–loss class partition, a normal-form game, and population-level observables under a specified adaptation rule. This integrated perspective is intended for mathematical social sciences: it separates strategic form from contextual valuation while preserving enough structure to compare invariance, equilibrium, welfare, and population behavior across domains.

The paper makes three principal contributions.

- (i) **Strategic contribution.** A general 2×2 theorem characterizes coexistence of intelligent and malicious pure equilibria. A shared-row configuration is impossible; a shared-column configuration forces indifference and is a knife-edge; and robust welfare-dominated coexistence is possible in a diagonal configuration under strict inequalities. An explicit game realizes the robust case with a certified open-ball radius.
- (ii) **Structural contribution.** The benefit–loss plane induces a partition of each finite profile space. We characterize class-preserving translations, define a Hamming pseudometric for contextual sensitivity, and give payoff- and weight-space criteria for robustness and boundary crossing. These results distinguish robustness of a profile’s class from robustness of its equilibrium status.
- (iii) **Computational contribution.** We give a reproducible exponential-replicator illustration and two audits: exhaustive enumeration of a finite grid of 2×2 games and seeded continuous payoff perturbations. The audits test the theorem implementation, finite-horizon convergence, and the separation between class and equilibrium robustness; they are not empirical validation.

The benchmark examples use three domains: a business organization with boss–employee players, a classroom with teacher–student players, and a sport team with coach–team members. The examples are not empirical measurements. They are normalized benchmark valuations generated from interpretable action features and context-specific weights. They show both stable profiles and context-sensitive profiles. In particular, the same abstract strategic profile—public pressure received with engaged uptake—may be malicious in a business organization, destructive in a classroom, and intelligent in a sport team. A second profile—constructive guidance met with resistance—shows a different transition pattern, from destructive to hapless to destructive.

The paper is a framework paper with a closed formal-computational contribution: it does not claim empirical validation of the numerical payoffs in the benchmark examples. However, it does not stop at definitions. The mathematical layer is complemented by equilibrium analysis and a reproducible minimal simulation. These layers show how contextual valuations can be used in strategic and population-based models without assuming an empirical calibration layer.

2 Related work and positioning

2.1 Cipolla-inspired games and simulations

Cipolla’s essay introduced a two-dimensional benefit–loss representation of social action. The core insight is that an action may benefit the actor, harm the other, benefit both, harm both, or benefit the other at the actor’s expense. This qualitative plane gives a natural bridge to game theory, since each interaction can be read as a payoff vector.

Several formal extensions have already appeared. Kuperman, Bárcenas and Kuperman proposed an evolutionary game inspired by Cipolla, extending classical cooperator–defector schemes by introducing additional strategies, including one associated with destructive behavior [12]. Bárcenas, Kuperman and Kuperman later studied a related evolutionary model in which destructive behavior can be adopted with a probability that affects the stationary distribution of strategies [2]. Mella proposed the Social Wheel as a didactic extension of Cipolla’s test by adding a third dimension related to the agent’s willingness to generate advantages for others [14]. Perissi and Bardi proposed a biophysical interpretation, connecting Cipolla’s ideas with predator–prey and resource-exploitation models [18]. Agent-based modeling has also been used to test Cipolla-like laws in artificial populations [25].

The present paper differs from these lines in three respects. First, it does not primarily study population dynamics or evolutionary equilibria, although it includes a minimal dynamic illustration. Second, it does not attempt to refine the psychological taxonomy of persons. Third, it treats context as a formal component of the model. The same strategic form may be interpreted by different valuation maps, and the mathematical problem is to determine which class properties are invariant under contextual translation.

2.2 Mathematical social sciences, welfare, and formal social modeling

The present paper is closest in spirit to the mathematical social sciences: it uses elementary formal structures to compare social meanings, payoff classifications, and strategic consequences across domains. We position it relative to four established literatures.

Social choice and welfare. Comparing strategic profiles by their induced classes and by the aggregate $W = x + y$ is, at bottom, a question of how individual gains and losses are aggregated into a social judgment. The classical theory of social choice and welfare studies exactly such aggregations and their structural limits [1, 23], while utility and preference-representation theory clarifies when numerical payoffs can stand in for underlying preferences [7]. Our W is a deliberately minimal aggregate used only as a diagnostic, not as a contribution to welfare theory.

Payoff signs and strategic incentives. The benefit–loss class of a profile depends only on the signs of its two payoff coordinates, whereas pure best responses depend on within-player payoff orderings [28, 16]. Mixed equilibria and the population update used below also depend on payoff differences, so they are not invariant under arbitrary ordinal transformations. The framework therefore keeps three objects separate: payoff signs for class, unilateral comparisons for pure Nash equilibrium, and cardinal payoff gaps for mixed and dynamic analysis. The benchmarks belong to the long tradition of 2×2 interaction games [20]; unlike games with vector payoffs, (x, y) here is the ordinary pair of scalar payoffs to two players, not a multi-criterion payoff for one player.

Equilibrium robustness. Stability and regularity of equilibrium components under game perturbations have an established technical meaning [10, 13]. We do not propose an equilibrium refinement or a notion of strategic stability in that sense. Proposition 10.6 instead asks a

narrower finite-game question: how large can a coordinate-wise payoff perturbation be while preserving the *complete set of pure Nash profiles exactly*? For 2×2 games the answer is an explicit radius in unilateral payoff slacks, which we compare with the independent radius preserving the benefit–loss class map.

Population and learning dynamics. The dynamic layer places the same valuations inside a discrete exponential-replicator, or multiplicative-weights, process, connecting the framework to evolutionary learning and adaptive game play [29, 9, 6, 22]. We use this layer only as a minimal aggregate illustration, not as an evolutionary or behavioral identification result.

Mathematical sociology and formal institutional models. Finally, treating context as a formal component that fixes how actions are valued is consistent with the tradition of compact formal models in mathematical sociology and institutional analysis [8, 4, 15, 17], where rules and roles determine the social meaning and consequences of behavior. In our terminology, an institution is encoded by a valuation map.

The local welfare function $W(x, y) = x + y$ used below is deliberately simple, but it allows the paper to separate two questions that are often conflated: whether a profile is a Nash equilibrium and whether it is socially desirable once both actor-side and affected-side payoffs are considered. This is important for the benchmark analysis, where one Nash equilibrium is malicious and welfare-dominated by another equilibrium.

2.3 Behavioral and contextual game theory

Behavioral game theory and decision theory have long emphasized that payoffs are not merely material outcomes. Framing, reference points, fairness, reciprocity, choice sets, and social preferences can alter evaluation of the same objective situation [11, 26, 27, 19, 5]. The present model is consistent with that view but does not identify preferences from observed choice: the contextual valuation V_C is an explicit model input. The paper provides a structural language for comparing consequences of alternative valuations, not a theory of how those valuations are formed.

Two recent contributions in *Mathematical Social Sciences* sharpen this boundary. Su and Morsky add norm-derived relational utility to canonical games and study resulting changes in cooperation, coordination, and equilibria [24]. Ryan permits preferences in a stochastic-choice model to depend on the category through which alternatives are considered [21]. Our object is different from both: the strategy set and normal form are held fixed, context enters as an exogenously specified valuation map on profiles, and the analysis compares the induced sign partition with the exact pure-equilibrium set under payoff perturbations. We therefore neither derive context from a social-norm utility component nor estimate it from stochastic choice.

2.4 Institutional analysis and social meaning

Institutional analysis emphasizes that rules, roles, and collective goals shape the meaning and consequences of actions [17]. A public correction has different institutional implications in a classroom, a firm, and a sport team because the relation between authority, performance, learning, discipline, and dignity differs across domains. In our terminology, institutional structure changes the valuation map. This is why the same strategic profile can be recontextualized without changing its abstract form.

2.5 Positioning

The paper therefore occupies a middle position within formal mathematical social modeling. It is not a psychological theory of stupidity and not an empirical study. It is a framework

for context-dependent benefit–loss interactions in which (i) classes are induced by contextual valuation maps, (ii) translations can preserve or fail to preserve those classes, (iii) the same valuations generate normal-form games, and (iv) simple population dynamics can be built from them.

3 The benefit–loss plane as a partition

Let $v = (x, y) \in \mathbb{R}^2$ be a payoff vector. The coordinate x measures the payoff of the acting player, and y measures the payoff of the affected player or group. Positive values represent gains, negative values represent losses, and zero values represent neutrality.

Definition 3.1 (benefit–loss regions). *The benefit–loss regions of the benefit–loss plane are*

$$\begin{aligned}\mathcal{I} &= \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0\}, \\ \mathcal{H} &= \{(x, y) \in \mathbb{R}^2 : x < 0, y > 0\}, \\ \mathcal{M} &= \{(x, y) \in \mathbb{R}^2 : x > 0, y < 0\}, \\ \mathcal{D}^- &= \{(x, y) \in \mathbb{R}^2 : x < 0, y < 0\}, \\ \mathcal{D}^0 &= \{(x, y) \in \mathbb{R}^2 : x = 0, y < 0\}.\end{aligned}$$

The destructive region is

$$\mathcal{D} = \mathcal{D}^- \cup \mathcal{D}^0.$$

The boundary region is

$$\partial\mathcal{C} = \mathbb{R}^2 \setminus (\mathcal{I} \cup \mathcal{H} \cup \mathcal{M} \cup \mathcal{D}).$$

The class $\mathcal{H}$ is called *hapless* rather than *helpless* in the sequel. It represents profiles in which the actor loses while the affected side gains. This terminology avoids suggesting a psychological incapacity and keeps the formal reading close to a payoff relation.

The inclusion of $\mathcal{D}^0$ in the destructive region (Figure 1) follows Cipolla’s idea that an action may harm the other while producing no benefit for the actor. The strict region $\mathcal{D}^-$ represents mutual loss, while $\mathcal{D}^0$ represents non-compensating harm with actor-neutral payoff. We keep this distinction explicit because $\mathcal{D}^0$ lies on an axis and is therefore classifiable but structurally non-robust. This is a modeling convention rather than a mathematical necessity: one could instead put all zero-coordinate cases into the boundary, but doing so would lose the actor-neutral case of non-compensating harm that motivates $\mathcal{D}^0$. The main preservation, sensitivity, and robustness results are formulated so that alternative boundary conventions can be adopted with only local changes to the class map Q . In particular, all structural results below remain valid under alternative boundary conventions, after replacing Q accordingly.

Definition 3.2 (benefit–loss class map). *Let*

$$\text{Class} = \{\text{I}, \text{H}, \text{M}, \text{D}, \partial\},$$

where I denotes intelligent, H hapless, M malicious, D destructive, and ∂ boundary. The class map

$$Q : \mathbb{R}^2 \longrightarrow \text{Class}$$

is defined by

$$Q(v) = \begin{cases} \text{I}, & v \in \mathcal{I}, \\ \text{H}, & v \in \mathcal{H}, \\ \text{M}, & v \in \mathcal{M}, \\ \text{D}, & v \in \mathcal{D}, \\ \partial, & v \in \partial\mathcal{C}. \end{cases}$$

Remark 3.3 (Actions rather than persons). The formal model classifies strategic profiles or interactions, not persons. A person may participate in different contexts and may generate profiles with different classes. This distinction is essential for avoiding psychological essentialism and for making the model suitable for institutional and educational analysis.

Definition 3.4 (Local welfare). *The local welfare associated with a payoff vector $v = (x, y)$ is*

$$W(v) = x + y.$$

Remark 3.5 (Welfare as a minimal diagnostic). We use $W = x + y$ only as a minimal diagnostic aggregate, not as a full welfare theory. It assumes interpersonal comparability and linear aggregation of actor-side and affected-side payoffs. We adopt it solely to separate, in the benchmarks, whether a profile is a Nash equilibrium from whether it is socially desirable once both sides are counted; richer or non-linear social welfare functionals could replace it without affecting the class-level results.

Proposition 3.6 (Destructive profiles have negative local welfare). *If $Q(x, y) = D$, then $W(x, y) < 0$.*

Proof. If $Q(x, y) = D$, then either $(x, y) \in \mathcal{D}^-$ or $(x, y) \in \mathcal{D}^0$. In the first case, $x < 0$ and $y < 0$, hence $x + y < 0$. In the second case, $x = 0$ and $y < 0$, hence again $x + y = y < 0$. $\square$

Remark 3.7 (Maliciousness and welfare transfer). Malicious profiles need not have negative local welfare. If $(x, y) = (5, -1)$, then the actor gains more than the affected player loses and $W > 0$. If $(x, y) = (1, -5)$, then the profile is still malicious, but $W < 0$. Thus aggregate welfare alone may hide relational harm. This mirrors the distinction in Cipolla's macro-level discussion between mere transfers of welfare and actions that destroy welfare.

Proposition 3.8 (Non-robustness of the actor-neutral destructive axis). *Every point $v = (0, y) \in \mathcal{D}^0$ with $y < 0$ is classifiable as destructive but is not locally class-robust. More precisely, for every $\varepsilon > 0$ there are points $v_M \in \mathcal{M}$ and $v_D \in \mathcal{D}^-$ with $\|v_M - v\|_\infty < \varepsilon$ and $\|v_D - v\|_\infty < \varepsilon$.*

Proof. Let $v = (0, y)$ with $y < 0$, and fix an arbitrary radius $\varepsilon > 0$. Since $y < 0$, the positive number $|y|/2$ is well defined. Choose

$$0 < \delta < \min\{\varepsilon, |y|/2\}.$$

Define

$$v_M = (\delta, y), \quad v_D = (-\delta, y).$$

The point v_M satisfies $\delta > 0$ and $y < 0$, hence $v_M \in \mathcal{M}$. The point v_D satisfies $-\delta < 0$ and $y < 0$, hence $v_D \in \mathcal{D}^-$. Moreover,

$$\|v_M - v\|_\infty = \max\{|\delta - 0|, |y - y|\} = \delta < \varepsilon,$$

and similarly

$$\|v_D - v\|_\infty = \max\{|-\delta - 0|, |y - y|\} = \delta < \varepsilon.$$

Thus every sup-norm neighborhood of v contains both malicious points and strictly destructive points. Although v is classified as destructive by convention, its class is not stable under arbitrarily small actor-side perturbations. $\square$

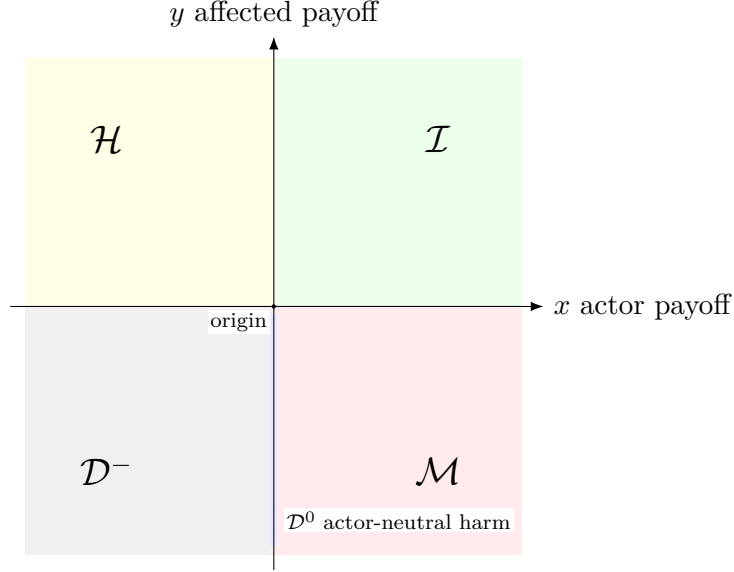


Figure 1: The benefit–loss partition of payoff space into the intelligent region $\mathcal{I}$, the hapless region $\mathcal{H}$, the malicious region $\mathcal{M}$, and the destructive region $\mathcal{D} = \mathcal{D}^- \cup \mathcal{D}^0$. The thin vertical strip is the actor-neutral harmful set $\mathcal{D}^0$, included in $\mathcal{D}$ by modeling convention.

4 Contextual benefit–loss games and induced partitions

A strategic form alone does not determine a benefit–loss class. The class depends on how a context evaluates the consequences of the strategies. We therefore attach a valuation map to each context.

Definition 4.1 (Contextual benefit–loss game). *A contextual benefit–loss game is a tuple*

$$G_C = (C, A, B, V_C),$$

where C is a context, A is the strategy set of the acting or authority-side player, B is the strategy set of the affected-side player, and

$$V_C : A \times B \longrightarrow \mathbb{R}^2$$

is a contextual valuation map. The profile set is denoted by

$$\Sigma_C = A \times B.$$

For $\sigma \in \Sigma_C$, the induced class is

$$Q_C(\sigma) = Q(V_C(\sigma)).$$

A context may represent an organization, a classroom, a sport team, a family, a public institution, or any domain in which actions generate gains and losses for multiple agents. The context is not merely a label: it determines which consequences count as gains or losses and how they are normalized.

Proposition 4.2 (One valuation, three induced layers). *Let $G_C = (C, A, B, V_C)$ be a contextual benefit–loss game and write $V_C(a, b) = (X_C(a, b), Y_C(a, b))$. Then the single valuation map $V_C : A \times B \rightarrow \mathbb{R}^2$ induces the following three structures.*

- (i) A class map $Q_C = Q \circ V_C : A \times B \rightarrow \text{Class}$ and hence a partition Π_C of the profile set.

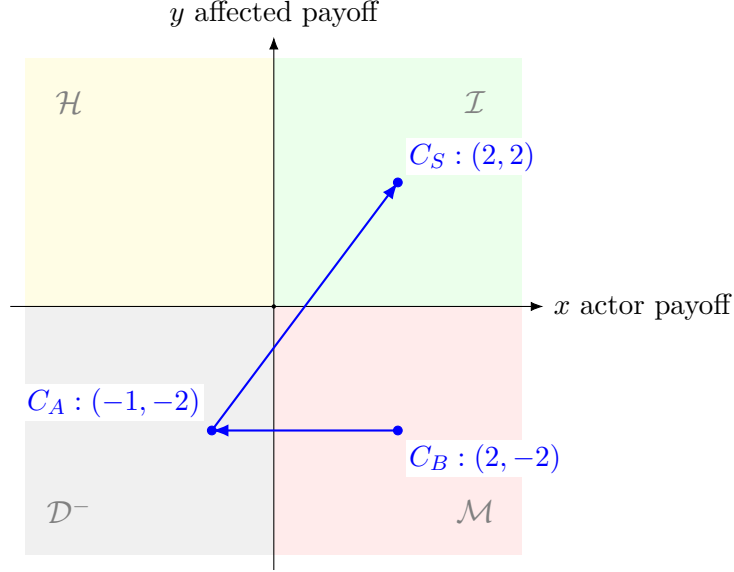


Figure 2: Benchmark recontextualization of the single abstract profile (P, U) across the three contexts: malicious in business C_B , destructive in classroom C_A , and intelligent in sport team C_S . The arrows represent contextual revaluation of the same strategic profile, not temporal motion.

- (ii) A two-player normal-form game with payoff functions $X_C : A \times B \rightarrow \mathbb{R}$ and $Y_C : A \times B \rightarrow \mathbb{R}$.
- (iii) Once an adaptation rule $\mathcal{L}$ is fixed, a population update map on mixed strategy states; in the exponential-replicator case of Section 11, this is a map $(p_t, q_t) \mapsto (p_{t+1}, q_{t+1})$.

Thus the classification, strategic, and population-dynamic layers are not separate inputs: they are different readings of the same contextual valuation.

Proof. For (i), the class map is obtained by composing the valuation V_C with the fixed benefit–loss class map $Q : \mathbb{R}^2 \rightarrow \text{Class}$. The fibers of this composite map are exactly the cells of the induced partition Π_C . For (ii), the two coordinate projections $\pi_1, \pi_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ give the payoff functions $X_C = \pi_1 \circ V_C$ and $Y_C = \pi_2 \circ V_C$, so the same profile set with these two payoff functions is a normal-form game. For (iii), any specified population update rule that uses these payoff functions as inputs determines a map on the chosen state space. In Section 11, the state is $(p_t, q_t) \in [0, 1]^2$ and the update is defined explicitly from the expected payoffs determined by X_C and Y_C . Hence all three layers are induced from V_C plus, for the dynamic layer, the chosen update rule. $\square$

Definition 4.3 (Induced benefit–loss partition). *Let $G_C = (C, A, B, V_C)$ be a contextual benefit–loss game. For each class $c \in \text{Class}$, define*

$$\Sigma_C^c = Q_C^{-1}(c) = \{\sigma \in A \times B : Q_C(\sigma) = c\}.$$

The induced benefit–loss partition of Σ_C is

$$\Pi_C = \{\Sigma_C^c : c \in \text{Class}\}.$$

Empty cells are allowed.

Proposition 4.4 (Classification is equivalent to partition membership). *For any profiles $\sigma, \tau \in \Sigma_C$,*

$$Q_C(\sigma) = Q_C(\tau)$$

if and only if σ and τ belong to the same cell of Π_C .

Proof. By definition, the cells of Π_C are exactly the inverse images $Q_C^{-1}(c)$ for $c \in \text{Class}$. Therefore two profiles have the same class precisely when they belong to the same inverse-image cell. $\square$

Definition 4.5 (Common strategic form). *Two contextual benefit-loss games $G_C = (C, A, B, V_C)$ and $G_D = (D, A, B, V_D)$ have a common strategic form if they share the same profile set $A \times B$ but may have different valuation maps.*

Definition 4.6 (Stable and recontextualized profile). *Let $\mathcal{C}$ be a family of contexts with a common strategic form. A profile $\sigma \in A \times B$ is stable across $\mathcal{C}$ if*

$$Q_C(\sigma) = Q_D(\sigma)$$

for all $C, D \in \mathcal{C}$. It is recontextualized between C and D if

$$Q_C(\sigma) \neq Q_D(\sigma).$$

Definition 4.7 (Payoff transition). *For two contexts C and D with common strategic form, the payoff transition of a profile σ is*

$$\Delta_{C \rightarrow D}(\sigma) = V_D(\sigma) - V_C(\sigma).$$

The associated class transition is

$$Q_C(\sigma) \longrightarrow Q_D(\sigma).$$

5 Class-preserving context translations

We now formalize the idea that one context can be translated into another. A translation may preserve the names of strategies, or it may map strategies in one context to analogous strategies in another context.

Definition 5.1 (Context translation). *Let $G_C = (C, A_C, B_C, V_C)$ and $G_D = (D, A_D, B_D, V_D)$ be contextual benefit-loss games. A context translation*

$$F : G_C \longrightarrow G_D$$

consists of two functions

$$F_A : A_C \rightarrow A_D, \quad F_B : B_C \rightarrow B_D.$$

The induced profile map is

$$F_* : A_C \times B_C \longrightarrow A_D \times B_D, \quad F_*(a, b) = (F_A(a), F_B(b)).$$

A translation is not automatically class-preserving. It simply tells us which strategies are considered analogous across contexts. Preservation depends on the valuations.

Definition 5.2 (Strong class-preserving translation). *A translation $F : G_C \rightarrow G_D$ is strong class-preserving if, for every profile $\sigma \in \Sigma_C$,*

$$Q_D(F_*\sigma) = Q_C(\sigma).$$

Theorem 5.3 (Partition characterization of strong preservation). *A translation $F : G_C \rightarrow G_D$ is strong class-preserving if and only if, for every $c \in \text{Class}$,*

$$\Sigma_C^c = F_*^{-1}(\Sigma_D^c).$$

Equivalently,

$$Q_D \circ F_* = Q_C.$$

Proof. We prove both directions.

Assume first that F is strong class-preserving. Fix a class $c \in \text{Class}$. We must show equality of two subsets of Σ_C . Let $\sigma \in \Sigma_C$. By the definition of the fiber Σ_C^c ,

$$\sigma \in \Sigma_C^c \iff Q_C(\sigma) = c.$$

Because F is strong class-preserving, $Q_C(\sigma) = Q_D(F_*\sigma)$ for every σ . Hence

$$Q_C(\sigma) = c \iff Q_D(F_*\sigma) = c.$$

The latter condition is equivalent to $F_*\sigma \in \Sigma_D^c$, which by definition of inverse image is equivalent to $\sigma \in F_*^{-1}(\Sigma_D^c)$. Thus, for arbitrary $\sigma \in \Sigma_C$,

$$\sigma \in \Sigma_C^c \iff \sigma \in F_*^{-1}(\Sigma_D^c),$$

so $\Sigma_C^c = F_*^{-1}(\Sigma_D^c)$.

Conversely, suppose that $\Sigma_C^c = F_*^{-1}(\Sigma_D^c)$ for every $c \in \text{Class}$. Let $\sigma \in \Sigma_C$ be arbitrary and set $c = Q_C(\sigma)$. Then $\sigma \in \Sigma_C^c$. By the assumed equality of fibers, $\sigma \in F_*^{-1}(\Sigma_D^c)$, hence $F_*\sigma \in \Sigma_D^c$. Therefore $Q_D(F_*\sigma) = c$. Since $c = Q_C(\sigma)$, we obtain

$$Q_D(F_*\sigma) = Q_C(\sigma).$$

As σ was arbitrary, F is strong class-preserving.

Finally, the identity $Q_D \circ F_* = Q_C$ is exactly the pointwise condition $Q_D(F_*\sigma) = Q_C(\sigma)$ for all $\sigma \in \Sigma_C$. Thus it is equivalent to strong class preservation. $\square$

Definition 5.4 (benefit–loss sign). *For $v = (x, y) \in \mathbb{R}^2$, define*

$$\text{csign}(v) = (\text{sgn } x, \text{sgn } y) \in \{-1, 0, 1\}^2.$$

A translation $F : G_C \rightarrow G_D$ is sign-preserving if

$$\text{csign}(V_D(F_*\sigma)) = \text{csign}(V_C(\sigma))$$

for every $\sigma \in \Sigma_C$.

Theorem 5.5 (Sign preservation implies class preservation). *Every sign-preserving translation is strong class-preserving.*

Proof. The class map Q is constant on the sign cells induced by the benefit–loss partition, with the convention that $(0, -)$ belongs to the destructive class and the remaining zero-coordinate cases belong to the boundary class. If F is sign-preserving, then $V_C(\sigma)$ and $V_D(F_*\sigma)$ have the same benefit–loss sign for every profile σ . Therefore they lie in the same benefit–loss region and have the same class. Hence $Q_D(F_*\sigma) = Q_C(\sigma)$ for every σ . $\square$

Definition 5.6 (Positive coordinate rescaling). *A map $R : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a positive coordinate rescaling if*

$$R(x, y) = (\alpha x, \beta y)$$

for some $\alpha, \beta > 0$.

Corollary 5.7 (Invariance under positive rescaling). *If $V_D(F_*\sigma) = R(V_C(\sigma))$ for all profiles σ , where R is a positive coordinate rescaling, then F is strong class-preserving.*

Proof. Positive coordinate rescaling preserves the benefit–loss sign of every payoff vector. The conclusion follows from the previous theorem. $\square$

Definition 5.8 (Weak welfare-preserving translation). *A translation $F : G_C \rightarrow G_D$ is weak welfare-preserving if, for every profile $\sigma \in \Sigma_C$,*

$$\text{sgn}(W(V_D(F_*\sigma))) = \text{sgn}(W(V_C(\sigma))).$$

Proposition 5.9 (Weak welfare preservation does not imply class preservation). *There exist contextual games and a translation that preserves the sign of local welfare but does not preserve benefit–loss class.*

Proof. Consider one-profile games G_C and G_D , and let F be the unique translation between them. Suppose

$$V_C(\sigma) = (2, -1), \quad V_D(F_*\sigma) = (1, 1).$$

Then $W(V_C(\sigma)) = 1$ and $W(V_D(F_*\sigma)) = 2$, so the sign of local welfare is positive in both games. Hence F is weak welfare-preserving. However, $Q_C(\sigma) = \mathbf{M}$ while $Q_D(F_*\sigma) = \mathbf{I}$. Thus F is not strong class-preserving. $\square$

The commuting condition is compositional. Identity translations preserve class, and if $Q_D \circ F_* = Q_C$ and $Q_E \circ G_* = Q_D$, then $Q_E \circ (G_* \circ F_*) = Q_C$. No additional categorical machinery is needed for the results below.

6 Contextual sensitivity as a pseudometric

To compare contexts quantitatively, we define an index measuring the fraction of profiles whose class changes.

Definition 6.1 (Contextual sensitivity). *Let $G_C = (C, A, B, V_C)$ and $G_D = (D, A, B, V_D)$ have the same finite strategic form, and let $\Sigma = A \times B$. The contextual sensitivity of C relative to D is*

$$S(C, D) = \frac{1}{|\Sigma|} |\{\sigma \in \Sigma : Q_C(\sigma) \neq Q_D(\sigma)\}|.$$

Equivalently, $S(C, D)$ is the normalized Hamming distance between the class maps $Q_C, Q_D : \Sigma \rightarrow \text{Class}$.

Definition 6.2 (Class-equivalence of contexts). *Two contexts with the same strategic form are class-equivalent, written $C \sim D$, if $Q_C = Q_D$ as functions $\Sigma \rightarrow \text{Class}$. Equivalently, $\Pi_C = \Pi_D$.*

Theorem 6.3 (Pseudometric property). *On any family of contexts with a common finite strategic form, S is a pseudometric. That is, for all contexts C, D, E ,*

- (a) $S(C, D) \geq 0$;
- (b) $S(C, D) = S(D, C)$;
- (c) $S(C, E) \leq S(C, D) + S(D, E)$;
- (d) $S(C, C) = 0$.

Moreover, $S(C, D) = 0$ if and only if $C \sim D$. Thus S descends to a metric on class-equivalence classes of contexts.

Proof. Let $\Sigma = A \times B$ be the common finite profile set. For two contexts C and D , write

$$\Delta(C, D) = \{\sigma \in \Sigma : Q_C(\sigma) \neq Q_D(\sigma)\}.$$

Then $S(C, D) = |\Delta(C, D)|/|\Sigma|$.

Nonnegativity follows because cardinalities are nonnegative. Also $\Delta(C, C) = \emptyset$, so $S(C, C) = 0$. Symmetry follows from the logical equivalence

$$Q_C(\sigma) \neq Q_D(\sigma) \Leftrightarrow Q_D(\sigma) \neq Q_C(\sigma),$$

which gives $\Delta(C, D) = \Delta(D, C)$ and hence $S(C, D) = S(D, C)$.

For the triangle inequality, fix $\sigma \in \Sigma$. Suppose $Q_C(\sigma) \neq Q_E(\sigma)$. If both $Q_C(\sigma) = Q_D(\sigma)$ and $Q_D(\sigma) = Q_E(\sigma)$ held, transitivity of equality would imply $Q_C(\sigma) = Q_E(\sigma)$, a contradiction. Hence at least one of the two inequalities $Q_C(\sigma) \neq Q_D(\sigma)$ or $Q_D(\sigma) \neq Q_E(\sigma)$ must hold. In indicator notation,

$$\mathbf{1}_{\{Q_C(\sigma) \neq Q_E(\sigma)\}} \leq \mathbf{1}_{\{Q_C(\sigma) \neq Q_D(\sigma)\}} + \mathbf{1}_{\{Q_D(\sigma) \neq Q_E(\sigma)\}}.$$

Summing this inequality over all $\sigma \in \Sigma$ and dividing by $|\Sigma|$ gives

$$S(C, E) \leq S(C, D) + S(D, E).$$

Finally, $S(C, D) = 0$ if and only if $|\Delta(C, D)| = 0$, which is equivalent to $\Delta(C, D) = \emptyset$, which is equivalent to $Q_C(\sigma) = Q_D(\sigma)$ for every $\sigma \in \Sigma$. This is precisely $Q_C = Q_D$, i.e. $C \sim D$. On the quotient by $\sim$, the only possible failure of identity of indiscernibles is removed, so the induced function is a metric. $\square$

Remark 6.4. The index S measures class sensitivity, not payoff distance. Two contexts may have very different numerical payoffs but the same induced partition, producing $S(C, D) = 0$. Conversely, a small numerical perturbation near an axis may change class.

7 Robustness and boundary crossing

The previous section measures whether profiles change class across contexts. We now give geometric criteria for when class changes are possible.

Definition 7.1 (Axis margin). For $v = (x, y) \in \mathbb{R}^2$, define its axis margin by

$$m(v) = \min\{|x|, |y|\}.$$

For a contextual game G_C , define

$$m_C(\sigma) = m(V_C(\sigma)).$$

A profile is axis-regular in context C if $m_C(\sigma) > 0$.

Axis-regular profiles do not lie on either coordinate axis. Profiles in $\mathcal{D}^0$ are classified as destructive but are not axis-regular, because they lie on the actor-neutral axis.

Theorem 7.2 (Pointwise robustness of class). Let G_C and G_D have a common strategic form. If σ is axis-regular in C and

$$\|V_D(\sigma) - V_C(\sigma)\|_\infty < m_C(\sigma),$$

then

$$Q_D(\sigma) = Q_C(\sigma).$$

Proof. Write $V_C(\sigma) = (x_C, y_C)$ and $V_D(\sigma) = (x_D, y_D)$. Since σ is axis-regular in C ,

$$m_C(\sigma) = \min\{|x_C|, |y_C|\} > 0.$$

In particular, $x_C \neq 0$ and $y_C \neq 0$, so $V_C(\sigma)$ lies in one of the four open sign quadrants. This excludes both the boundary region and the actor-neutral destructive axis $\mathcal{D}^0$.

The sup-norm inequality gives

$$|x_D - x_C| \leq \|V_D(\sigma) - V_C(\sigma)\|_\infty < m_C(\sigma) \leq |x_C|,$$

and likewise $|y_D - y_C| < |y_C|$. If $x_C > 0$, then $x_D > x_C - |x_D - x_C| > 0$. If $x_C < 0$, then $x_D < x_C + |x_D - x_C| < 0$. Thus $\text{sgn}(x_D) = \text{sgn}(x_C)$. The same argument gives $\text{sgn}(y_D) = \text{sgn}(y_C)$.

Therefore $V_C(\sigma)$ and $V_D(\sigma)$ remain in the same open sign quadrant. On open sign quadrants the class map Q is constant: the four quadrants are exactly $\mathcal{I}$, $\mathcal{H}$, $\mathcal{M}$, and $\mathcal{D}^-$. Hence $Q(V_D(\sigma)) = Q(V_C(\sigma))$, i.e. $Q_D(\sigma) = Q_C(\sigma)$. $\square$

Corollary 7.3 (Uniform robustness). *Let G_C and G_D have a common finite strategic form Σ . Suppose every profile is axis-regular in C , and set*

$$\delta_C = \min_{\sigma \in \Sigma} m_C(\sigma) > 0.$$

If

$$\sup_{\sigma \in \Sigma} \|V_D(\sigma) - V_C(\sigma)\|_\infty < \delta_C,$$

then $S(C, D) = 0$.

Proof. The stated uniform bound implies the pointwise hypothesis of the previous theorem for every $\sigma \in \Sigma$. Hence no profile changes class, so $S(C, D) = 0$. $\square$

Definition 7.4 (Axis set). *The axis set of the benefit-loss partition is*

$$\mathcal{A} = \{(x, y) \in \mathbb{R}^2 : x = 0 \text{ or } y = 0\}.$$

Theorem 7.5 (Axis-crossing criterion). *Let G_C and G_D have a common strategic form. If*

$$Q_C(\sigma) \neq Q_D(\sigma),$$

then the closed line segment joining $V_C(\sigma)$ and $V_D(\sigma)$ intersects the axis set $\mathcal{A}$.

Proof. Let $V_C(\sigma) = (x_C, y_C)$ and $V_D(\sigma) = (x_D, y_D)$. The closed segment between them is the image of

$$\gamma : [0, 1] \rightarrow \mathbb{R}^2, \quad \gamma(t) = (1 - t)V_C(\sigma) + tV_D(\sigma).$$

Write $\gamma(t) = (x(t), y(t))$, where $x(t) = (1 - t)x_C + tx_D$ and $y(t) = (1 - t)y_C + ty_D$. Suppose, for contradiction, that $\gamma([0, 1]) \cap \mathcal{A} = \emptyset$. Then $x(t) \neq 0$ and $y(t) \neq 0$ for all $t \in [0, 1]$. Since $x(t)$ is continuous on the connected interval $[0, 1]$ and never vanishes, its sign is constant on the interval. Hence $x_C = x(0)$ and $x_D = x(1)$ have the same sign. The same argument applied to $y(t)$ shows that y_C and y_D have the same sign.

Thus $V_C(\sigma)$ and $V_D(\sigma)$ lie in the same open sign quadrant. If an endpoint lay on an axis, the segment would intersect $\mathcal{A}$, contrary to the assumption. Hence no endpoint lies on an axis, and $Q_C(\sigma) = Q_D(\sigma)$, contradicting the hypothesis. Therefore the segment must intersect $\mathcal{A}$. $\square$

Remark 7.6. Because $\mathcal{D}^0 \subset \mathcal{D}$ lies on an axis, not every axis intersection is a class change. The theorem is one-way: class change implies axis crossing. Its value is geometric: recontextualization cannot occur by small drift entirely inside a single open sign cell.

Proposition 7.7 (Axis crossing is not sufficient for class change). *There exist payoff vectors $v, w \in \mathbb{R}^2$ such that the segment $[v, w]$ intersects $\mathcal{A}$ but $Q(v) = Q(w)$.*

Proof. Take $v = (-1, -1)$ and $w = (0, -1)$. The segment joining them lies on the horizontal line $y = -1$ and contains $w = (0, -1)$, so it intersects the axis set $\mathcal{A}$. However, $v \in \mathcal{D}^-$ and $w \in \mathcal{D}^0$, and both regions are included in the destructive class $\mathcal{D}$. Hence $Q(v) = Q(w) = \mathcal{D}$. Thus axis crossing is necessary but not sufficient for class change. $\square$

8 Feature-based contextual valuation

The benchmark payoffs used below are generated by a transparent feature-based parameterization. This representation makes every numerical input inspectable and provides a template for later calibration, but it does not identify the weights empirically or make the synthetic payoffs non-arbitrary.

Definition 8.1 (Feature-based valuation). *Let $\Sigma = A \times B$ be a finite profile set. A feature representation is a map*

$$\phi : \Sigma \rightarrow \mathbb{R}^k.$$

A context-specific weight matrix is a matrix

$$W_C \in \mathbb{R}^{2 \times k}.$$

The induced contextual valuation is

$$V_C(\sigma) = W_C \phi(\sigma).$$

The first row of W_C determines the actor-side payoff, and the second row determines the affected-side payoff.

The same feature vector may be evaluated differently in different contexts. For example, public pressure may increase short-term compliance in a business organization, damage pedagogical legitimacy in a classroom, and improve focus in a sport team when embedded in a shared competitive culture.

Proposition 8.2 (Feature weights induce contextual revaluation). *Let $\phi : \Sigma \rightarrow \mathbb{R}^k$ be fixed. If $W_C \neq W_D$, then the same profile σ may satisfy*

$$Q(W_C \phi(\sigma)) \neq Q(W_D \phi(\sigma)).$$

Proof. It suffices to give an example. Let $k = 1$, $\phi(\sigma) = 1$, $W_C = (1, -1)^T$ and $W_D = (1, 1)^T$. Then $W_C \phi(\sigma) = (1, -1)$ is malicious, while $W_D \phi(\sigma) = (1, 1)$ is intelligent. Hence changing the context weights can change the class while the profile feature vector is fixed. $\square$

For the benchmark examples we use two abstract strategies for the authority-side player:

$$G = \text{constructive guidance}, \quad P = \text{public pressure},$$

and two abstract strategies for the affected-side player:

$$U = \text{engaged uptake}, \quad R = \text{resistance or withdrawal}.$$

The symbol U is used for uptake in order to avoid confusion with the destructive class $\mathcal{D}$.

The feature vector is

$$\phi(\sigma) = (g, p, u, r, pu, gr, pr)^T \in \mathbb{R}^7,$$

Feature	Interpretation
g	Baseline effect of constructive guidance by the authority-side player.
p	Baseline effect of public pressure by the authority-side player.
u	Baseline effect of engaged uptake by the affected-side player.
r	Baseline effect of resistance or withdrawal by the affected-side player.
pu	Contextual interaction effect when public pressure is met with engaged uptake.
gr	Contextual interaction effect when guidance is met with resistance.
pr	Contextual interaction effect when pressure is met with resistance.

Table 1: Interpretation of the benchmark features. The weights assigned to these features encode context-specific norms, goals, and role expectations.

Profile	g	p	u	r	pu	gr	pr
(G, U)	1	0	1	0	0	0	0
(G, R)	1	0	0	1	0	1	0
(P, U)	0	1	1	0	1	0	0
(P, R)	0	1	0	1	0	0	1

Table 2: Feature representation for the four benchmark profiles.

where g, p, u, r are indicators for guidance, pressure, uptake, and resistance, while pu, gr, pr are interaction indicators for (P, U) , (G, R) , and (P, R) respectively. Table 1 explains the features. Table 2 gives the four feature vectors.

The weights in Table 3 are illustrative. Their signs encode the intended institutional readings: in the business benchmark, pressure can increase short-term actor-side control but reduce the affected side’s autonomy and trust; in the classroom benchmark it is penalized on both sides; and in the sport benchmark it is positive only when combined with uptake. These statements define the examples and are not empirical claims about the three domains.

The representation is deliberately saturated relative to the four observed profiles: seven features generate only four payoff vectors per context, so the weight matrix is not identified from those vectors. Multiple weight matrices produce the same game. Accordingly, the feature layer is an auditable parameterization rather than an explanatory regression model. The formal robustness claims below concern sign-defined sets of payoffs or weights; empirical identification would require additional profiles, restrictions, or external data.

Proposition 8.3 (Class regions in weight space). *Fix a profile σ and its feature vector $\phi(\sigma) \in \mathbb{R}^k$. For each open class $c \in \{\mathbf{I}, \mathbf{H}, \mathbf{M}\}$, the set of weights $W \in \mathbb{R}^{2 \times k}$ such that $Q(W\phi(\sigma)) = c$ is an intersection of two strict linear half-spaces. The strict destructive region $\mathcal{D}^-$ is also an intersection of two strict linear half-spaces. Thus, away from $\mathcal{D}^0$ and the boundary, each class condition is stable under sufficiently small perturbations of W .*

Proof. Write the two rows of W as $w_1, w_2 \in \mathbb{R}^k$. Then

$$W\phi(\sigma) = (w_1 \cdot \phi(\sigma), w_2 \cdot \phi(\sigma)).$$

For fixed $\phi(\sigma)$, the maps $W \mapsto w_1 \cdot \phi(\sigma)$ and $W \mapsto w_2 \cdot \phi(\sigma)$ are linear functions of the entries of W . The class conditions are therefore linear sign conditions on these two functions. Explicitly,

$$Q(W\phi(\sigma)) = \mathbf{I} \iff w_1 \cdot \phi(\sigma) > 0 \text{ and } w_2 \cdot \phi(\sigma) > 0,$$

$$Q(W\phi(\sigma)) = \mathbf{H} \iff w_1 \cdot \phi(\sigma) < 0 \text{ and } w_2 \cdot \phi(\sigma) > 0,$$

$$Q(W\phi(\sigma)) = \mathbf{M} \iff w_1 \cdot \phi(\sigma) > 0 \text{ and } w_2 \cdot \phi(\sigma) < 0,$$

Context	Payoff row	g	p	u	r	pu	gr	pr
C_B business	actor	1	0	1	-2	1	0	0
	affected	1	-1	2	-2	-3	0	0
C_A classroom	actor	1	-2	1	-3	0	0	3
	affected	1	-4	2	0	0	0	1
C_S sport team	actor	2	1	1	-3	0	0	0
	affected	1	0	2	-2	0	0	-1

Table 3: Context-specific weight matrices. Applying $V_C(\sigma) = W_C\phi(\sigma)$ generates the benchmark payoff matrices in Table 6.

and

$$W\phi(\sigma) \in \mathcal{D}^- \iff w_1 \cdot \phi(\sigma) < 0 \text{ and } w_2 \cdot \phi(\sigma) < 0.$$

Each condition is the intersection of two strict linear half-spaces in $\mathbb{R}^{2 \times k}$. Strict linear half-spaces are open sets, and finite intersections of open sets are open. Hence if a weight matrix W places $W\phi(\sigma)$ in one of these open classes, there exists a radius $\rho > 0$ such that every weight matrix W' with $\|W' - W\| < \rho$ induces the same class for σ . The exceptional cases are precisely those in which one of the two linear forms is zero, corresponding to $\mathcal{D}^0$ or to the boundary of the benefit–loss partition. $\square$

This proposition addresses sensitivity, not empirical validity. The benchmark tables are one point in weight space, and nearby weights preserving the induced signs leave the class conclusions unchanged. This does not establish that the selected region describes any real institution.

8.1 Robustness of benchmark transitions in weight space

The next result is the main formal response to the concern that the benchmark payoffs are hand-crafted. It shows that the reported class transitions are not isolated numerical coincidences: they persist on open polyhedral regions of the weight space. Thus the exact entries in Table 3 are illustrative representatives of broader sign-defined parameter regions.

Proposition 8.4 (Open weight conditions for the benchmark transitions). *Let $\phi_{PU} = \phi(P, U)$ and $\phi_{GR} = \phi(G, R)$. Write the rows of the business, classroom, and sport-team weight matrices as*

$$W_B = (w_B^A, w_B^B), \quad W_A = (w_A^A, w_A^B), \quad W_S = (w_S^A, w_S^B),$$

where the superscripts A and B denote actor-side and affected-side rows. Then the transition

$$(P, U) : \mathbf{M} \longrightarrow \mathbf{D} \longrightarrow \mathbf{I}$$

holds, with the classroom point in $\mathcal{D}^-$, if and only if

$$w_B^A \cdot \phi_{PU} > 0, \quad w_B^B \cdot \phi_{PU} < 0,$$

$$w_A^A \cdot \phi_{PU} < 0, \quad w_A^B \cdot \phi_{PU} < 0,$$

and

$$w_S^A \cdot \phi_{PU} > 0, \quad w_S^B \cdot \phi_{PU} > 0.$$

Similarly,

$$(G, R) : \mathbf{D} \longrightarrow \mathbf{H} \longrightarrow \mathbf{D}$$

with the destructive points in $\mathcal{D}^-$ holds if and only if

$$w_B^A \cdot \phi_{GR} < 0, \quad w_B^B \cdot \phi_{GR} < 0,$$

$$w_A^A \cdot \phi_{GR} < 0, \quad w_A^B \cdot \phi_{GR} > 0,$$

and

$$w_S^A \cdot \phi_{GR} < 0, \quad w_S^B \cdot \phi_{GR} < 0.$$

Each set of inequalities defines an open polyhedral region in the product of the three weight spaces.

Proof. For any context C , the class of a profile is determined by the signs of the two coordinates of $W_C\phi(\sigma)$, namely

$$W_C\phi(\sigma) = (w_C^A \cdot \phi(\sigma), w_C^B \cdot \phi(\sigma)).$$

For (P, U) to be malicious in the business context, these two signs must be $(+, -)$, giving the first pair of inequalities. For (P, U) to be strictly destructive in the classroom context, the signs must be $(-, -)$, giving the second pair. For (P, U) to be intelligent in the sport-team context, the signs must be $(+, +)$, giving the third pair. This proves the first equivalence.

The proof for (G, R) is identical. The transition $D \rightarrow H \rightarrow D$ with strict destructive endpoints requires signs $(-, -)$ in business, $(-, +)$ in classroom, and $(-, -)$ in sport. Translating these sign requirements into dot-product inequalities gives the displayed conditions.

Finally, each displayed condition is a strict linear inequality in the entries of the corresponding weight row. A finite conjunction of strict linear inequalities is an open polyhedral region. Therefore the benchmark transitions are not isolated numerical coincidences; they persist throughout open regions of the relevant weight spaces. $\square$

The inequalities also identify which signs are substantively essential. For example, the transition $(P, U) : M \rightarrow D \rightarrow I$ requires public pressure with uptake to be actor-beneficial and affected-side harmful in the business context, harmful to both sides in the classroom context, and beneficial to both sides in the sport-team context. If any of these sign requirements is reversed, the transition changes. The feature weights should therefore be read as a compact representation of institutional norms and role expectations rather than as empirical estimates.

Profile	Business C_B	Classroom C_A	Sport team C_S	Class transition
(P, U)	$(+, -)$	$(-, -)$	$(+, +)$	$M \rightarrow D \rightarrow I$
(G, R)	$(-, -)$	$(-, +)$	$(-, -)$	$D \rightarrow H \rightarrow D$

Table 4: Essential sign patterns behind the two recontextualized benchmark profiles. The exact numerical payoffs in Table 6 are representatives of these sign-defined regions.

Context	Sign of (P, U)	Institutional reading of the essential signs
Business C_B	$(+, -)$	Public pressure met with uptake raises the actor’s short-term control but erodes the affected side’s autonomy and trust.
Classroom C_A	$(-, -)$	Public pressure damages pedagogical legitimacy and student confidence, harming both sides.
Sport team C_S	$(+, +)$	Public pressure accepted as tactical discipline benefits both the actor and the team.

Table 5: Institutional reading of the essential sign conditions for the recontextualized profile (P, U) . It is the signs, not the exact magnitudes, that carry the social interpretation; this is exactly what Proposition 8.4 formalizes.

9 Benchmark contextual games

This section introduces three benchmark examples showing that the same strategic form may induce different benefit–loss classifications when evaluated under different contexts. The examples are normalized payoff models designed to make explicit the role of contextual valuation.

We consider three contexts:

$$C_B = \text{business organization}, \quad C_A = \text{classroom}, \quad C_S = \text{sport team}.$$

In each context there is an authority-side player and an affected-side player:

$$C_B : \text{boss–employee}, \quad C_A : \text{teacher–student}, \quad C_S : \text{coach–team members}.$$

9.1 Business organization: boss–employee

In the business context, the first payoff measures managerial control, efficiency, reputation, or short-term organizational advantage. The second payoff measures the employee’s autonomy, motivation, professional growth, and work climate. Applying the business weights W_{C_B} gives

	U : engaged uptake	R : resistance/withdrawal
G : constructive guidance	(2, 3) [I]	(−1, −1) [D]
P : public pressure	(2, −2) [M]	(−2, −3) [D]

The profile (G, U) is intelligent: constructive guidance improves the manager’s objective and benefits the employee. The profile (P, U) is classified as malicious: the boss obtains compliance, control, or short-term efficiency, while the employee loses autonomy, confidence, or trust. The profile (P, R) is destructive for both sides.

9.2 Classroom: teacher–student

In the classroom context, the first payoff measures the teacher’s pedagogical effectiveness, legitimacy, and classroom climate. The second payoff measures the student’s learning, confidence, participation, and epistemic autonomy. Applying the classroom weights W_{C_A} gives

	U : engaged uptake	R : resistance/withdrawal
G : constructive guidance	(2, 3) [I]	(−2, 1) [H]
P : public pressure	(−1, −2) [D]	(−2, −3) [D]

The profile (G, U) remains intelligent. However, (G, R) becomes hapless: the teacher invests time, patience, and emotional effort, while the student still obtains some benefit from the pedagogical support. In contrast, public pressure is unstable in the classroom. Even when the student apparently complies, the interaction may damage confidence, participation, and classroom climate. Thus (P, U) is classified as destructive, not malicious, because the teacher also loses pedagogical legitimacy.

9.3 Sport team: coach–team members

In the sport context, the first payoff measures the coach’s tactical effectiveness, authority, and match-oriented performance. The second payoff measures the team’s cohesion, discipline, motivation, and competitive performance. Applying the sport-team weights W_{C_S} gives

	U : engaged uptake	R : resistance/withdrawal
G : constructive guidance	(3, 3) [I]	(−1, −1) [D]
P : public pressure	(2, 2) [I]	(−2, −3) [D]

In this context, (P, U) may become intelligent. Public tactical pressure, when embedded in a shared competitive culture and received with disciplined uptake, can increase focus, cohesion, and collective performance. This is not a claim that public pressure is always beneficial in sport settings; it depends on timing, trust, norms, and the distinction between tactical urgency and humiliation. The same strategy remains destructive when it is received as humiliation, unfair blame, or excessive pressure. Hence (P, R) remains destructive.

Strategic profile	Business C_B	Classroom C_A	Sport team C_S
(G, U)	$(2, 3)$ [I]	$(2, 3)$ [I]	$(3, 3)$ [I]
(G, R)	$(-1, -1)$ [D]	$(-2, 1)$ [H]	$(-1, -1)$ [D]
(P, U)	$(2, -2)$ [M]	$(-1, -2)$ [D]	$(2, 2)$ [I]
(P, R)	$(-2, -3)$ [D]	$(-2, -3)$ [D]	$(-2, -3)$ [D]

Table 6: Contextual payoff transitions for the same abstract strategic profiles.

9.4 Payoff and class transitions

Table 6 summarizes the three contextual valuations. For the profile (P, U) ,

$$V_{C_B}(P, U) = (2, -2), \quad V_{C_A}(P, U) = (-1, -2), \quad V_{C_S}(P, U) = (2, 2).$$

Therefore,

$$M \longrightarrow D \longrightarrow I.$$

This is the central phenomenon of recontextualization: the same abstract profile may move from malicious, to destructive, to intelligent depending on the contextual valuation (Figure 2).

The second profile (G, R) gives a different transition:

$$D \longrightarrow H \longrightarrow D.$$

Thus recontextualization need not be a movement toward a better class. It can also change the interpretation of loss: in the classroom, teacher effort under student resistance may still help the student, while in the other contexts the same abstract profile is destructive for both sides.

For the profile (G, U) ,

$$I \longrightarrow I \longrightarrow I.$$

This profile is structurally stable and intelligent across the three contexts. For the profile (P, R) ,

$$D \longrightarrow D \longrightarrow D.$$

This profile is structurally stable, but destructively so.

Proposition 9.1 (Stability and recontextualization in the benchmark). *For the three benchmark contexts:*

- (a) (G, U) is stable and intelligent;
- (b) (P, R) is stable and destructive;
- (c) (P, U) is recontextualized, with transition $M \rightarrow D \rightarrow I$;
- (d) (G, R) is recontextualized, with transition $D \rightarrow H \rightarrow D$.

Proof. We compute the class of each profile from the signs of its payoff vector in Table 6.

For (G, U) , the three payoff vectors are $(2, 3)$ in C_B , $(2, 3)$ in C_A , and $(3, 3)$ in C_S . Each has positive first coordinate and positive second coordinate. Hence all three are in $\mathcal{I}$, so the profile is stable and intelligent.

For (P, R) , the three payoff vectors are all $(-2, -3)$. Both coordinates are negative in each context, so each vector lies in $\mathcal{D}^- \subset \mathcal{D}$. Hence the profile is stable and destructive.

For (P, U) , the business payoff is $(2, -2)$, whose signs are $(+, -)$, so the class is M. The classroom payoff is $(-1, -2)$, whose signs are $(-, -)$, so the class is D. The sport-team payoff is $(2, 2)$, whose signs are $(+, +)$, so the class is I. Thus the transition is $M \rightarrow D \rightarrow I$.

For (G, R) , the business payoff $(-1, -1)$ is destructive, the classroom payoff $(-2, 1)$ is hapless because the actor-side coordinate is negative and the affected-side coordinate is positive, and the sport-team payoff $(-1, -1)$ is destructive. Hence the transition is $D \rightarrow H \rightarrow D$. $\square$

Proposition 9.2 (Sensitivity of the benchmark contexts). *For the three benchmark contexts,*

$$S(C_B, C_A) = \frac{2}{4}, \quad S(C_B, C_S) = \frac{1}{4}, \quad S(C_A, C_S) = \frac{2}{4}.$$

Proof. The common profile set has four elements:

$$\Sigma = \{(G, U), (G, R), (P, U), (P, R)\}.$$

By the induced class maps and by Table 6, the classes in C_B and C_A differ exactly on (G, R) and (P, U) . They agree on (G, U) and (P, R) . Therefore the number of changed profiles is 2, and

$$S(C_B, C_A) = \frac{2}{|\Sigma|} = \frac{2}{4}.$$

Between C_B and C_S , only (P, U) changes class: (G, U) remains intelligent, (G, R) remains destructive, and (P, R) remains destructive. Thus $S(C_B, C_S) = 1/4$. Between C_A and C_S , the profiles (G, R) and (P, U) change class, while (G, U) and (P, R) do not. Hence $S(C_A, C_S) = 2/4$. $\square$

Profile	m_{C_B}	m_{C_A}	m_{C_S}
(G, U)	2	2	3
(G, R)	1	1	1
(P, U)	2	1	2
(P, R)	2	2	2

Table 7: Axis margins for the benchmark profiles. Larger margins indicate greater local robustness of class under payoff perturbations.

The axis margins in Table 7 show how the robustness theorem applies. For example, the classroom payoff $V_{C_A}(P, U) = (-1, -2)$ has margin 1; perturbations of sup-norm smaller than 1 cannot change its class. Thus the benchmark conclusions are not destroyed by arbitrarily small noise, although profiles with smaller margins are less robust.

10 Game-theoretic layer: best responses and equilibria

The partition-based layer classifies profiles by the signs of their contextual payoff vectors. Since each valuation also gives a normal-form game, it is useful to separate two notions that are sometimes conflated: *class invariance* and *Nash equilibrium membership*. A profile can be intelligent without being a Nash equilibrium, and a malicious profile can be a Nash equilibrium if no player has a profitable unilateral deviation.

Definition 10.1 (Class desirability). *A profile σ in a context C is benefit–loss desirable (equivalently, class-desirable) if $Q_C(\sigma) = 1$, that is, if both the actor side and the affected side gain. Benefit–loss desirability is thus a purely class-based property of a single profile. It is distinct from welfare dominance, which compares the aggregate $W = x + y$ of two profiles, and from Nash equilibrium membership, which concerns unilateral deviations. Whenever we contrast desirability with equilibrium below, “desirable” means class 1.*

Let $G_C = (C, A, B, V_C)$ be a contextual benefit–loss game and write

$$V_C(a, b) = (X_C(a, b), Y_C(a, b)),$$

where X_C is the payoff of the acting player and Y_C is the payoff of the affected player. The best-response correspondences are

$$\text{BR}_C^A(b) = \arg \max_{a \in A} X_C(a, b), \quad \text{BR}_C^B(a) = \arg \max_{b \in B} Y_C(a, b).$$

A pure Nash equilibrium is a profile (a, b) such that

$$a \in \text{BR}_C^A(b), \quad b \in \text{BR}_C^B(a).$$

Definition 10.2 (Order-preserving context translation). *A translation $F : G_C \rightarrow G_D$ is order-preserving if for every $b \in B_C$ and all $a, a' \in A_C$,*

$$X_C(a, b) \geq X_C(a', b) \iff X_D(F_A a, F_B b) \geq X_D(F_A a', F_B b),$$

and for every $a \in A_C$ and all $b, b' \in B_C$,

$$Y_C(a, b) \geq Y_C(a, b') \iff Y_D(F_A a, F_B b) \geq Y_D(F_A a, F_B b').$$

Proposition 10.3 (Order preservation implies equilibrium preservation). *Let $F : G_C \rightarrow G_D$ be a bijective order-preserving translation. If (a, b) is a pure Nash equilibrium of G_C , then $F_*(a, b)$ is a pure Nash equilibrium of G_D .*

Proof. Since (a, b) is a Nash equilibrium in G_C , a maximizes $X_C(\cdot, b)$ and b maximizes $Y_C(a, \cdot)$. Order preservation implies that $F_A a$ maximizes $X_D(\cdot, F_B b)$ among the images of row strategies, and that $F_B b$ maximizes $Y_D(F_A a, \cdot)$ among the images of column strategies. Since F is bijective on both strategy sets, these images exhaust the strategy sets of G_D . Hence $F_*(a, b)$ is a pure Nash equilibrium of G_D . $\square$

Remark 10.4. Class preservation and equilibrium preservation are logically distinct. Class preservation depends on signs of payoff vectors. Equilibrium preservation depends on ordinal comparisons of payoffs within each player’s unilateral deviations. A translation may preserve one structure while failing to preserve the other.

Context	$\text{BR}^A(U)$	$\text{BR}^A(R)$	$\text{BR}^B(G)$	$\text{BR}^B(P)$	Pure NE	Classes
C_B	$\{G, P\}$	$\{G\}$	$\{U\}$	$\{U\}$	$(G, U), (P, U)$	I, M
C_A	$\{G\}$	$\{G, P\}$	$\{U\}$	$\{U\}$	(G, U)	I
C_S	$\{G\}$	$\{G\}$	$\{U\}$	$\{U\}$	(G, U)	I

Table 8: Best-response correspondences and pure Nash equilibria in the benchmark contexts.

Proposition 10.5 (Benchmark equilibria). *The pure Nash equilibria of the benchmark games are exactly those reported in Table 8. In particular, (P, U) is a malicious Nash equilibrium in the business context, while (P, U) is intelligent but not a Nash equilibrium in the sport context.*

Proof. We verify each context by best responses. In C_B , when the affected side plays U , the acting-side payoffs are $X(G, U) = 2$ and $X(P, U) = 2$, so both G and P are best responses. When the affected side plays R , $X(G, R) = -1 > X(P, R) = -2$, so G is the unique best response. For the affected side, $Y(G, U) = 3 > Y(G, R) = -1$ and $Y(P, U) = -2 > Y(P, R) = -3$, so U is always the unique best response. Therefore (G, U) and (P, U) are the pure equilibria. Their classes are I and M.

In C_A , against U the acting side prefers G because $2 > -1$; against R it is indifferent because $-2 = -2$. The affected side always prefers U , since $3 > 1$ against G and $-2 > -3$ against P . Hence the only intersection of best responses is (G, U) , of class I.

In C_S , the acting side always prefers G , since $3 > 2$ against U and $-1 > -2$ against R . The affected side always prefers U , since $3 > -1$ against G and $2 > -3$ against P . Hence the unique pure equilibrium is (G, U) , of class I. Although (P, U) is intelligent in C_S , the acting side has a profitable unilateral deviation to G , so it is not an equilibrium. $\square$

This result is one of the main conceptual payoffs of the game-theoretic layer. Benefit–loss desirability and equilibrium membership are different dimensions. The business benchmark contains a malicious equilibrium: the profile admits no profitable unilateral deviation but is relationally harmful to the affected side. Conversely, the sport benchmark contains an intelligent non-equilibrium: the profile benefits both sides but is dominated for the acting side by constructive guidance. The equilibrium membership of (P, U) in C_B rests on the exact actor indifference $X_{C_B}(G, U) = X_{C_B}(P, U)$; Proposition 11.2 shows that this equilibrium is a knife-edge in payoff magnitude, in contrast to the open-region robustness of its malicious *class*.

The distinction can be quantified exactly. For a profile (i, j) in a 2×2 game, let i' and j' denote the alternative actions and define the unilateral advantages

$$a_{ij} = X(i, j) - X(i', j), \quad b_{ij} = Y(i, j) - Y(i, j').$$

Thus (i, j) is a pure Nash equilibrium exactly when $a_{ij}, b_{ij} \geq 0$.

Proposition 10.6 (Exact robustness radius of the pure-equilibrium set). *For each profile (i, j) define*

$$r_{ij} = \begin{cases} \frac{1}{2} \min\{a_{ij}, b_{ij}\}, & (i, j) \in \text{NE}(G), \\ \frac{1}{2} \max(\{-a_{ij} : a_{ij} < 0\} \cup \{-b_{ij} : b_{ij} < 0\}), & (i, j) \notin \text{NE}(G). \end{cases}$$

Then

$$\rho_{\text{NE}}(G) = \min_{i,j} r_{ij}$$

is the supremal open sup-norm radius that preserves the complete set of pure Nash equilibria: every payoff perturbation of norm less than $\rho_{\text{NE}}(G)$ leaves $\text{NE}(G)$ unchanged, while for every $\varepsilon > \rho_{\text{NE}}(G)$ some perturbation of norm less than ε changes it.

Proof. Each unilateral advantage is a difference of two payoff coordinates, so a sup-norm perturbation of radius δ changes it by less than 2δ . For an equilibrium profile, both advantages therefore remain nonnegative whenever $\delta < \frac{1}{2} \min\{a_{ij}, b_{ij}\}$. For a non-equilibrium profile, at least one advantage is negative; it remains non-equilibrium whenever one negative advantage remains negative, which is guaranteed for δ below half the largest negative slack. Taking the minimum over all four profiles preserves the entire equilibrium set.

For sharpness, choose a profile attaining the minimum. If it is an equilibrium, perturb the payoff coordinate associated with a smallest advantage downward and its deviation payoff upward; any radius strictly above r_{ij} makes that advantage negative. If it is not an equilibrium, raise the current payoff and lower the deviation payoff for every negative advantage; because the two players' payoff coordinates are distinct, a perturbation of radius r_{ij} makes all such advantages nonnegative. In either case arbitrarily small enlargement beyond $\rho_{\text{NE}}(G)$ changes equilibrium membership. $\square$

Let $\rho_Q(G) = \min_{\sigma} m(V(\sigma))$ be the uniform class-map radius from Corollary 7.3. Direct substitution gives Table 9. The business benchmark has $\rho_Q = 1$ but $\rho_{NE} = 0$: its class map is locally constant on a full neighborhood, whereas its equilibrium set is not constant on any neighborhood. The other two benchmarks have positive, but smaller, equilibrium-set radii.

Context	Class-map radius ρ_Q	Pure-NE-set radius ρ_{NE}
Business C_B	1	0
Classroom C_A	1	1/2
Sport team C_S	1	1/2

Table 9: Exact open sup-norm robustness radii for the benchmark class maps and complete pure-equilibrium sets.

Proposition 10.7 (Class desirability and Nash equilibrium membership are independent). *In the benchmark family, benefit–loss desirability does not imply Nash equilibrium membership, and Nash equilibrium membership does not imply benefit–loss desirability.*

Proof. First, consider the sport-team context C_S . Table 6 gives $V_{C_S}(P, U) = (2, 2)$, so (P, U) is intelligent. However, Table 8 and Proposition 10.5 show that the unique pure Nash equilibrium in C_S is (G, U) . Equivalently, when the affected side plays U , the acting side obtains 3 from G and 2 from P , so the acting side has a profitable unilateral deviation from (P, U) to (G, U) . Thus an intelligent profile need not be a Nash equilibrium.

Second, consider the business context C_B . Table 6 gives $V_{C_B}(P, U) = (2, -2)$, so (P, U) is malicious. Proposition 10.5 shows that (P, U) is nevertheless a pure Nash equilibrium in C_B : the acting side is indifferent between G and P against U , and the affected side strictly prefers U to R against P . Thus Nash equilibrium membership does not imply benefit–loss desirability. $\square$

Proposition 10.8 (Mixed equilibria in the benchmark). *In the benchmark family, the business game has a continuum of mixed Nash equilibria in which the affected-side player chooses U with probability one and the authority-side player uses any mixture over $\{G, P\}$. The classroom and sport-team games have the unique mixed Nash equilibrium concentrated on (G, U) .*

Proof. Let p denote the probability of G and q the probability of U . In the business context, the affected-side player strictly prefers U to R against both G and P , since $3 > -1$ and $-2 > -3$. Hence any mixed equilibrium must have $q = 1$. When $q = 1$, the authority-side payoffs from G and P are both equal to 2, so every mixture over G and P is a best response. Therefore every state (p, q) with $p \in [0, 1]$ and $q = 1$ is a mixed equilibrium.

In the classroom context, the affected-side player strictly prefers U to R against both authority-side strategies, since $3 > 1$ and $-2 > -3$. Hence $q = 1$ in equilibrium. Against $q = 1$, the authority-side player strictly prefers G to P , since $2 > -1$. Thus the unique equilibrium is $p = 1, q = 1$, i.e. (G, U) .

In the sport-team context, the affected-side player strictly prefers U to R against both authority-side strategies, since $3 > -1$ and $2 > -3$. The authority-side player strictly prefers G to P against both affected-side strategies, since $3 > 2$ and $-1 > -2$. Hence the unique equilibrium is again $p = 1, q = 1$, i.e. (G, U) . $\square$

Definition 10.9 (Pareto and welfare dominance). *For two profiles σ, τ in a fixed context C , write $V_C(\sigma) = (x_{\sigma}, y_{\sigma})$ and $V_C(\tau) = (x_{\tau}, y_{\tau})$. We say that σ Pareto-dominates τ if $x_{\sigma} \geq x_{\tau}$ and $y_{\sigma} \geq y_{\tau}$, with at least one strict inequality. We say that σ welfare-dominates τ if $W(V_C(\sigma)) > W(V_C(\tau))$.*

Proposition 10.10 (A malicious equilibrium can be welfare-dominated). *In the business benchmark, (P, U) is a malicious Nash equilibrium, but it is Pareto-dominated and welfare-dominated by the intelligent Nash equilibrium (G, U) .*

Proof. From Table 6, in the business context

$$V_{C_B}(G, U) = (2, 3), \quad V_{C_B}(P, U) = (2, -2).$$

The acting-side payoff is the same in the two profiles, namely 2. The affected-side payoff is strictly higher at (G, U) , since $3 > -2$. Hence (G, U) Pareto-dominates (P, U) . Moreover,

$$W(V_{C_B}(G, U)) = 2 + 3 = 5,$$

whereas

$$W(V_{C_B}(P, U)) = 2 + (-2) = 0.$$

Thus (G, U) welfare-dominates (P, U) . Proposition 10.5 shows that both profiles are Nash equilibria in C_B , while Table 6 gives classes I and M, respectively. Therefore a malicious equilibrium can be both Pareto-dominated and welfare-dominated by another equilibrium. $\square$

This proposition is the main welfare lesson of the benchmark. Nash equilibrium membership alone does not identify the socially preferable equilibrium: in C_B , both (G, U) and (P, U) are equilibria, but only the former is both intelligent and welfare superior.

10.1 A general characterization of welfare-dominated malicious equilibria

Proposition 10.10 is a statement about one benchmark. We now show that it is an instance of a general structural dichotomy for 2×2 contextual benefit-loss games. The dichotomy explains why the business malicious equilibrium is fragile and predicts a second, robust regime that the benchmarks do not exhibit. Throughout this subsection a *game* is identified with its four payoff vectors

$$v = (V(G, U), V(G, R), V(P, U), V(P, R)) \in \mathbb{R}^8,$$

and a property is called *robust* if it holds on an open neighborhood of v in $\mathbb{R}^8$.

Lemma 10.11 (Indifference forced by coexisting equilibria). *Let a 2×2 contextual benefit-loss game have two distinct pure Nash equilibria. If they share the same actor-side row, the affected-side player is indifferent between U and R in that row. If they share the same affected-side column, the actor-side player is indifferent between G and P in that column.*

Proof. Suppose (a, U) and (a, R) are both equilibria for a fixed row a . Affected-side optimality at (a, U) gives $Y_C(a, U) \geq Y_C(a, R)$, and at (a, R) gives $Y_C(a, R) \geq Y_C(a, U)$; hence $Y_C(a, U) = Y_C(a, R)$. The column case is symmetric, using actor-side optimality to obtain $X_C(G, b) = X_C(P, b)$ for the shared column b . $\square$

Theorem 10.12 (Configuration theorem for intelligent and malicious equilibria). *Let G_C be a 2×2 contextual benefit-loss game, let σ_I be an intelligent pure Nash equilibrium, and let $\sigma_M \neq \sigma_I$ be a malicious pure Nash equilibrium. Then exactly one of the following relative positions occurs, and it determines robustness as indicated.*

- (a) **(Shared row: impossible.)** σ_I and σ_M cannot share an actor-side row.
- (b) **(Shared column: forced knife-edge.)** If σ_I and σ_M share an affected-side column, then the actor is indifferent in that column, $X_C(\sigma_I) = X_C(\sigma_M) > 0$, and σ_I both Pareto-dominates and welfare-dominates σ_M . Every game in this configuration lies in the hyperplane $\{X_C(\sigma_I) = X_C(\sigma_M)\} \subset \mathbb{R}^8$, so the configuration has empty interior and is never robust.

- (c) (**Diagonal: the only possible robust regime.**) Otherwise σ_I and σ_M are diagonal. Diagonal position alone does not imply robustness. If both profiles are strict equilibria and $W(\sigma_I) > W(\sigma_M)$, their class, equilibrium, and welfare-dominance conditions are strict linear inequalities in v and therefore persist on an open neighborhood. Such strict diagonal instances exist.

In particular, diagonal position is necessary but not sufficient for robust coexistence of an intelligent equilibrium and a welfare-dominated malicious equilibrium. A shared-column instance is necessarily a knife-edge.

Proof. (a) If σ_I and σ_M shared a row, Lemma 10.11 would give $Y_C(\sigma_I) = Y_C(\sigma_M)$. But $\sigma_I \in \mathbf{I}$ forces $Y_C(\sigma_I) > 0$ and $\sigma_M \in \mathbf{M}$ forces $Y_C(\sigma_M) < 0$, a contradiction.

(b) If they share a column, Lemma 10.11 gives $X_C(\sigma_I) = X_C(\sigma_M)$. Both classes require a positive actor payoff, so this common value is positive. Since $Y_C(\sigma_I) > 0 > Y_C(\sigma_M)$ while the actor payoffs are equal, σ_I Pareto-dominates σ_M , and

$$W(\sigma_I) = X_C(\sigma_I) + Y_C(\sigma_I) > X_C(\sigma_M) + Y_C(\sigma_M) = W(\sigma_M).$$

The defining equality $X_C(\sigma_I) = X_C(\sigma_M)$ confines the configuration to a hyperplane of $\mathbb{R}^8$, which has empty interior; any perturbation breaking the equality removes one of the two equilibria, so the configuration is not robust.

(c) By (a) and (b) the only remaining relative position is diagonal. Weak diagonal equilibria can still rely on payoff ties, so diagonal position alone is not sufficient. When both equilibria are strict, however, their best-response conditions are strict linear inequalities. The sign conditions defining $\sigma_I \in \mathbf{I}$ and $\sigma_M \in \mathbf{M}$, together with welfare dominance $W(\sigma_I) - W(\sigma_M) > 0$, are also strict linear inequalities in the coordinates of v . A finite intersection of strict half-spaces is open. Proposition 10.14 supplies a nonempty instance, proving that a robust diagonal regime exists. $\square$

Corollary 10.13 (The business benchmark is a knife-edge instance). *In the business context, $\sigma_I = (G, U)$ and $\sigma_M = (P, U)$ share the affected-side action U , so Theorem 10.12(b) applies: the actor indifference $X_{C_B}(G, U) = X_{C_B}(P, U) = 2$ is forced, (G, U) Pareto- and welfare-dominates (P, U) , and the malicious equilibrium is non-robust. This recovers Propositions 10.10 and 11.2 as the shared-action case of the dichotomy.*

Proof. Both profiles use the column U , so the configuration is the shared-column case of Theorem 10.12(b). The forced indifference and the dominance are exactly Proposition 10.10, and the resulting non-robustness is Proposition 11.2. $\square$

Proposition 10.14 (Robust welfare-dominated malicious equilibria exist). *Let $\Gamma^\diamond$ be the 2×2 contextual benefit-loss game with payoff vectors*

$$V(G, U) = (1, 4), \quad V(G, R) = (-2, 2), \quad V(P, U) = (-1, -3), \quad V(P, R) = (3, -1).$$

Then (G, U) is an intelligent strict Nash equilibrium and (P, R) is a malicious strict Nash equilibrium; (G, U) welfare-dominates (P, R) but does not Pareto-dominate it; and all of these properties hold simultaneously for every payoff matrix at sup-norm distance strictly less than $3/4$ from $\Gamma^\diamond$. The bound is sharp for this certificate: at distance $3/4$ the welfare inequality can become an equality. Hence robust welfare-dominated malicious equilibria exist, realizing case (c) of Theorem 10.12.

Proof. Direct evaluation gives the strict best-response inequalities

$$X(G, U) - X(P, U) = 2 > 0, \quad Y(G, U) - Y(G, R) = 2 > 0,$$

so (G, U) is a strict equilibrium, and

$$X(P, R) - X(G, R) = 5 > 0, \quad Y(P, R) - Y(P, U) = 2 > 0,$$

so (P, R) is a strict equilibrium. The class conditions $X(G, U), Y(G, U) > 0$ and $X(P, R) > 0 > Y(P, R)$ hold, so $(G, U) \in \mathcal{I}$ and $(P, R) \in \mathcal{M}$. Welfare dominance holds because $W(G, U) - W(P, R) = 5 - 2 = 3 > 0$, while Pareto dominance fails because $X(P, R) - X(G, U) = 2 > 0$, so the malicious actor strictly out-earns the intelligent one. Each displayed quantity is a strict linear form $\ell(v)$ in the eight payoff coordinates; under a perturbation of sup-norm size δ it changes by at most $\delta \|\ell\|_1$, where $\|\ell\|_1$ is the sum of the absolute coefficients of ℓ . The smallest ratio $\ell(\Gamma^\diamond)/\|\ell\|_1$ over the list is attained at the welfare form, with slack 3 and $\|\ell\|_1 = 4$, giving $3/4$. Hence all inequalities remain valid for every $\delta < 3/4$. At $\delta = 3/4$, decreasing both coordinates of $V(G, U)$ and increasing both coordinates of $V(P, R)$ by $3/4$ makes the two welfare values equal, so the open-ball qualification is necessary. $\square$

Remark 10.15 (Robust dynamic survival of malicious behavior). The malicious equilibrium (P, R) of $\Gamma^\diamond$ is strict, so the vertex $(p, q) = (0, 0)$ is a locally asymptotically stable rest point of the exponential-replicator update of Section 11 for every $\eta > 0$. The Jacobian eigenvalues at that vertex are $\exp(-5\eta)$ and $\exp(-2\eta)$, obtained from the actor and affected selection gaps $X(G, R) - X(P, R) = -5$ and $Y(P, U) - Y(P, R) = -2$. Both lie strictly inside the unit circle. Hence a neighborhood of $(0, 0)$ is attracted to it and a strictly positive malicious mass survives without payoff indifference. The knife-edge of the business case is therefore a feature of its shared-action configuration, not a general law.

11 Minimal population-dynamics illustration

The previous section interprets each contextual valuation as a normal-form game. This section shows how the same valuations can feed a minimal aggregate population-dynamics model. This is not an agent-based validation or an empirical simulation of real organizations. Its purpose is to demonstrate that the framework can generate reproducible population-level quantities such as expected welfare, class frequencies, and trajectory distances.

11.1 Exponential-replicator update

Consider a large population of authority-side agents choosing between G and P , and affected-side agents choosing between U and R . Let p_t be the probability that an authority-side agent chooses G at iteration t , and let q_t be the probability that an affected-side agent chooses U . For a context C , define expected row payoffs

$$\begin{aligned} u_G(q_t) &= q_t X_C(G, U) + (1 - q_t) X_C(G, R), \\ u_P(q_t) &= q_t X_C(P, U) + (1 - q_t) X_C(P, R), \end{aligned}$$

and expected column payoffs

$$\begin{aligned} v_U(p_t) &= p_t Y_C(G, U) + (1 - p_t) Y_C(P, U), \\ v_R(p_t) &= p_t Y_C(G, R) + (1 - p_t) Y_C(P, R). \end{aligned}$$

We use the discrete exponential-replicator, or multiplicative-weights, update

$$\begin{aligned} p_{t+1} &= \frac{p_t \exp(\eta u_G(q_t))}{p_t \exp(\eta u_G(q_t)) + (1 - p_t) \exp(\eta u_P(q_t))}, \\ q_{t+1} &= \frac{q_t \exp(\eta v_U(p_t))}{q_t \exp(\eta v_U(p_t)) + (1 - q_t) \exp(\eta v_R(p_t))}. \end{aligned}$$

Here $\eta > 0$ is a learning-rate or payoff-sensitivity parameter. The rule adds η times the payoff gap to each population's log odds and is the two-action multiplicative-weights update used in adaptive game play [6]. It preserves the unit square and its faces but contains no mutation term or independent choice noise; it should therefore not be confused with logit quantal response. We use it as a deterministic comparative-dynamics device rather than as a behaviorally estimated learning model [9, 22].

The induced profile probabilities are

$$\begin{aligned}\pi_t(G, U) &= p_t q_t, & \pi_t(G, R) &= p_t(1 - q_t), \\ \pi_t(P, U) &= (1 - p_t)q_t, & \pi_t(P, R) &= (1 - p_t)(1 - q_t).\end{aligned}$$

From these probabilities we compute expected local welfare and class masses:

$$\begin{aligned}E_t[W] &= \sum_{\sigma \in \Sigma} \pi_t(\sigma) W(V_C(\sigma)), \\ \Pr_t(c) &= \sum_{\sigma \in \Sigma: Q_C(\sigma)=c} \pi_t(\sigma), \quad c \in \text{Class}.\end{aligned}$$

11.2 Illustrative trajectories

For an illustrative run we use $p_0 = q_0 = 0.5$, $\eta = 0.45$, and 50 iterations. The code uses exactly the payoff matrices in Table 6. Figure 3 separates the probability of constructive guidance from the probability of engaged uptake, improving readability relative to a single overlaid plot.

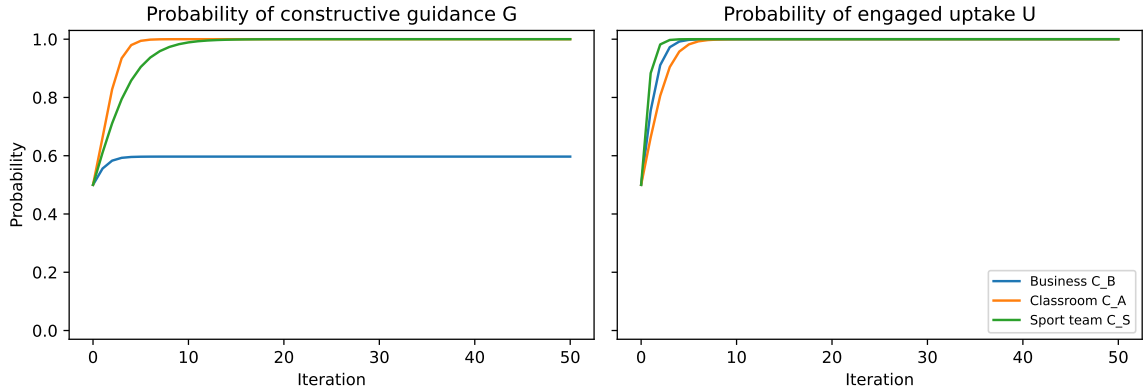


Figure 3: Discrete exponential-replicator trajectories for the benchmark contexts. Left: probability of constructive guidance G . Right: probability of engaged uptake U .

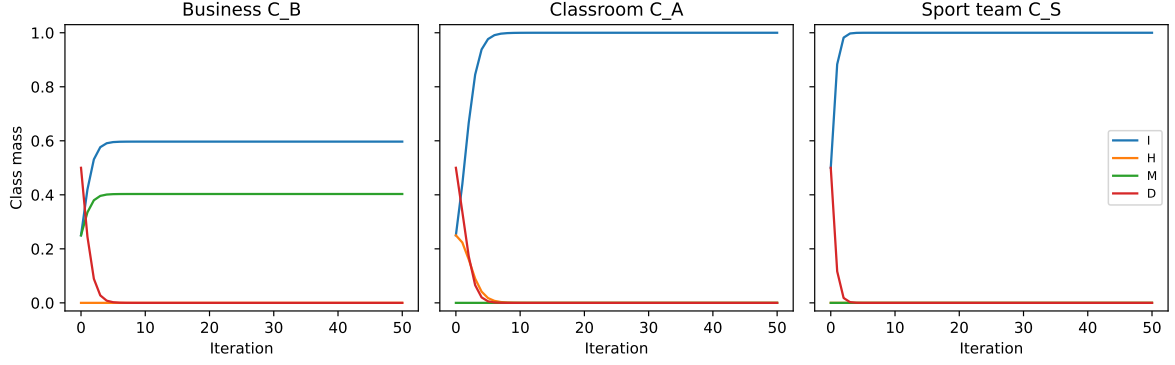


Figure 4: Class masses induced by the same exponential-replicator trajectories. The business context retains a positive malicious mass because (P, U) remains strategically viable under engaged uptake.

The simulation illustrates a point that is invisible in the static class table alone. The business context may converge to engaged uptake while retaining a positive mass on public pressure because public pressure and constructive guidance tie in the actor payoff against uptake. Hence some malicious class mass remains. In the classroom and sport contexts, the dynamics concentrate on constructive guidance and engaged uptake. This does not prove that real classrooms or sport teams behave this way; rather, it shows how contextual valuations can be used to generate testable population-level predictions once empirical calibration is available.

Context	p_{50}	q_{50}	$E_{50}[W]$	$\text{Pr}_{50}(I)$	$\text{Pr}_{50}(M)$	$\text{Pr}_{50}(D)$
Business C_B	0.597	1.000	2.985	0.597	0.403	0.000
Classroom C_A	1.000	1.000	5.000	1.000	0.000	0.000
Sport team C_S	1.000	1.000	6.000	1.000	0.000	0.000

Table 10: Final values for the illustrative run with $p_0 = q_0 = 0.5$, $\eta = 0.45$, and 50 iterations.

11.3 Fixed points, local stability, and the equilibrium knife-edge

The exponential-replicator update has enough structure to describe its fixed points and to explain why the business context behaves differently from the classroom and sport contexts. Write p for the probability of G and q for the probability of U , and recall the expected actor-side payoffs

$$u_G(q) = qX_C(G, U) + (1 - q)X_C(G, R), \quad u_P(q) = qX_C(P, U) + (1 - q)X_C(P, R).$$

Proposition 11.1 (Invariant faces and benchmark fixed points). *For every context C , each coordinate value in $\{0, 1\}$ is invariant under the exponential-replicator map: if $p_t \in \{0, 1\}$ then $p_{t+1} = p_t$, and likewise for q . Consequently the four vertices of $[0, 1]^2$ are fixed points, including vertices that are not Nash equilibria, and (G, U) , i.e. $(p, q) = (1, 1)$, is fixed in all three benchmark contexts. On the invariant edge $q = 1$ the dynamics of p are monotone and governed by the sign of*

$$u_G(1) - u_P(1) = X_C(G, U) - X_C(P, U).$$

Proof. If $p_t = 1$ the numerator and denominator of the p -update coincide, so $p_{t+1} = 1$; if $p_t = 0$ the numerator vanishes, so $p_{t+1} = 0$. The same argument applies to q . Hence the edges $\{p \in \{0, 1\}\}$ and $\{q \in \{0, 1\}\}$ are invariant, and their four intersection points are fixed. At $q = 1$ one has $u_G(1) = X_C(G, U)$ and $u_P(1) = X_C(P, U)$, and the exponential-replicator

p -update satisfies $p_{t+1} > p_t$ when $u_G(1) > u_P(1)$, $p_{t+1} < p_t$ when $u_G(1) < u_P(1)$, and $p_{t+1} = p_t$ when $u_G(1) = u_P(1)$. $\square$

The three benchmark contexts differ precisely in the actor-side selection gap along the uptake direction. A direct computation from Table 6 gives

$$u_G(q) - u_P(q) = \begin{cases} 1 - q, & \text{business } C_B, \\ 3q, & \text{classroom } C_A, \\ 1, & \text{sport } C_S. \end{cases}$$

Proposition 11.2 (Strategic and dynamic non-robustness of the malicious phenomenon). *In the business context the actor-side selection gap is $u_G(q) - u_P(q) = 1 - q$, which vanishes as $q \rightarrow 1$. Consequently the segment $\{(p, 1) : p \in [0, 1]\}$ is a continuum of fixed points, (P, U) is a Nash equilibrium, and every interior initial condition leaves a strictly positive malicious mass $1 - p_\infty > 0$ in the limit. These properties use the exact actor indifference $X_{C_B}(G, U) = X_{C_B}(P, U)$. More precisely, hold all other business payoffs fixed and replace $X_{C_B}(G, U) = 2$ by $2 + \varepsilon$. If $\varepsilon > 0$, the dynamics from every interior state converge to (G, U) and malicious mass vanishes; if $\varepsilon < 0$, they converge to (P, U) and malicious mass tends to one. Thus arbitrarily small one-coordinate perturbations select opposite limiting equilibria. By contrast, in the classroom and sport contexts the actor gap equals $3q$ and 1 , which are positive near $q = 1$, so (G, U) attracts every interior trajectory and malicious mass vanishes.*

Proof. The gap formulas follow by substituting the payoffs of Table 6 into $u_G(q) - u_P(q)$. In the unperturbed business game the gap is 0 at $q = 1$, so every p is fixed on that edge; it is exactly the continuum of mixed equilibria of Proposition 10.8. For an interior start, the affected-side gap is $v_U(p) - v_R(p) = 3p + 1 \geq 1$, so the log odds of q increase by at least η per step and $1 - q_t \rightarrow 0$ geometrically. The cumulative actor drift $\sum_t \eta(1 - q_t)$ is finite, hence the log odds of p_t remain finite and $p_t \rightarrow p_\infty \in (0, 1)$. Under the stated one-coordinate perturbation, the actor gap becomes $1 - q + \varepsilon q$. The affected-side equation is unchanged, so $q_t \rightarrow 1$. If $\varepsilon > 0$, the actor gap is eventually bounded below by a positive constant and the log odds of p_t diverge to $+\infty$, giving $p_t \rightarrow 1$. If $\varepsilon < 0$, the gap is eventually bounded above by a negative constant and the log odds diverge to $-\infty$, giving $p_t \rightarrow 0$. In the classroom and sport contexts, $q_t \rightarrow 1$ and the actor gaps $3q$ and 1 are eventually bounded away from zero, so $p_t \rightarrow 1$. $\square$

This proposition isolates the conceptual core of the business benchmark. The *class* of (P, U) is robust: it remains malicious on an open region of weight and payoff space. Coexistence of the intelligent and malicious equilibria is not robust because it lies on the indifference hyperplane $X_{C_B}(G, U) = X_{C_B}(P, U)$. Class robustness and equilibrium robustness are therefore different requirements; the perturbation audit below quantifies this distinction without treating the synthetic draws as empirical data.

11.4 Sensitivity to learning rate and initial conditions

A single run is not sufficient to support even a minimal population-dynamics claim. We therefore report a deterministic sensitivity check over the wider grid

$$\eta \in \{0.05, 0.15, 0.45, 1.00, 1.50\}$$

and the five initial conditions

$$(p_0, q_0) \in \{(0.2, 0.2), (0.2, 0.8), (0.5, 0.5), (0.8, 0.2), (0.8, 0.8)\}.$$

The resulting summary over 25 deterministic runs per context is shown in Table 11. The full grid is exported by the reproducibility notebook.

Context	p_{50}	q_{50}	$E_{50}[W]$	$\text{Pr}_{50}(\text{I})$	$\text{Pr}_{50}(\text{M})$	$\text{Pr}_{50}(\text{D})$
Business C_B	[.223, .932]	[.971, 1]	[1.094, 4.661]	[.222, .932]	[.068, .775]	[0, .029]
Classroom C_A	[.942, 1]	[.918, 1]	[4.059, 5]	[.864, 1]	[0, 0]	[0, .058]
Sport team C_S	[.753, 1]	[.9999, 1]	[5.505, 6]	[.9999, 1]	[0, 0]	[0, .0002]

Table 11: Ranges over the 25 deterministic design points formed by five learning rates and five initial conditions. These are design extrema, not sampling estimates or confidence intervals.

The zero malicious mass in the classroom and sport rows follows from their class maps: neither game has a malicious profile. In the business game, public pressure remains payoff-equivalent to constructive guidance against uptake, so positive malicious mass persists at every design point. Its magnitude depends on the initial state and learning rate; the claim is viability under this constructed game, not a unique or empirically estimated mass.

11.5 Computational robustness audits

Two additional audits challenge the claims rather than merely regenerate the chosen example. First, we enumerate all $5^8 = 390,625$ 2×2 games whose eight payoff coordinates lie in $\{-2, -1, 0, 1, 2\}$. Including zero admits boundary payoffs and additional weak best-response ties. Among these games there are 14,148 unordered pairs consisting of an intelligent and a malicious pure equilibrium. None shares an actor row, 5,400 share a column with zero failures of the dominance conclusion in Theorem 10.12(b), and 8,748 are diagonal. Of the diagonal pairs, 1,372 consist of strict equilibria with the intelligent profile welfare-dominating the malicious profile. This finite audit is not a proof, but it checks the implementation against every game in a nontrivial grid and searches explicitly for counterexamples.

Second, for each benchmark and each radius $\delta \in \{.25, .50, .75, .99, 1.01, 1.50\}$, we draw 2,000 independent perturbation vectors uniformly from $[-\delta, \delta]^8$ using the fixed seed 20260622. For every $\delta < 1$, all 6,000 perturbed games preserve their benchmark class maps, as predicted by the minimum axis margin. In the business game, however, none preserves the original two-equilibrium set: a continuous perturbation breaks the actor tie with probability one. Approximately half retain a malicious equilibrium alone, depending on the sign of the broken tie. Figure 5 contrasts class-map retention with equilibrium-set retention. The rates describe the stated synthetic perturbation distribution; they are not inferential estimates about real contexts.

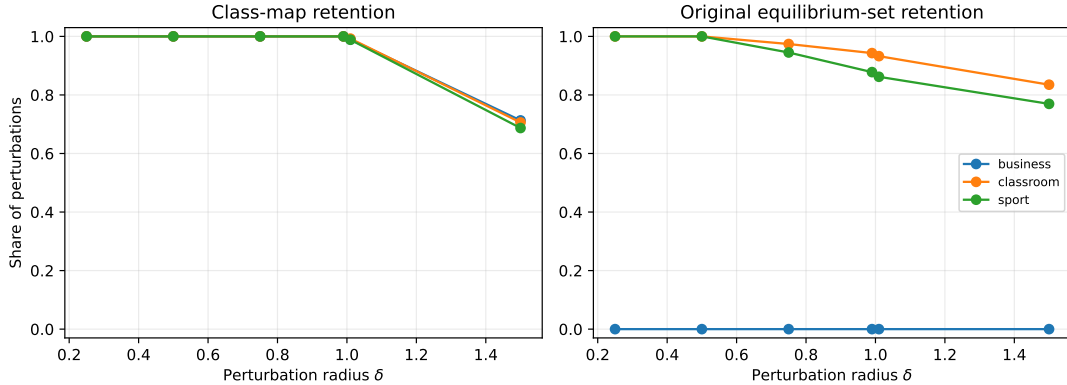


Figure 5: Retention rates under 2,000 seeded uniform payoff perturbations per context and radius. Class maps are locally robust, while the business benchmark’s original equilibrium set is destroyed by every nonzero continuous perturbation because it requires an exact tie.

11.6 Connecting class sensitivity with dynamic distance

The structural pseudometric S compares contexts by class maps. A population model also yields trajectory distributions π_t^C over profiles. For a fixed update rule L , horizon T , learning rate η , and initial condition $z_0 = (p_0, q_0)$, define

$$D_{\text{traj}}^{L,T,\eta,z_0}(C, D) = \frac{1}{T+1} \sum_{t=0}^T \frac{1}{2} \|\pi_t^C - \pi_t^D\|_1.$$

The factor $\frac{1}{2}$ makes each summand the total-variation distance between the two profile distributions, so $D_{\text{traj}} \in [0, 1]$ and is directly comparable with the structural class sensitivity $S(C, D) \in [0, 1]$. When no ambiguity is possible, we write D_{traj} . This notation emphasizes that the dynamic distance is not an intrinsic distance between contexts alone; it is relative to the chosen population process.

Proposition 11.3 (Dynamic distance is process-dependent). *For fixed update rule L , horizon T , learning rate η , and initial condition z_0 , $D_{\text{traj}}^{L,T,\eta,z_0}$ is a pseudometric on the set of contexts with a common strategic form. It depends on the induced trajectories and is not determined by the structural class sensitivity S alone.*

Proof. For each context C , the update rule determines a trajectory $(\pi_t^C)_{t=0}^T$ in the simplex over the common finite profile set. Nonnegativity and symmetry of $D_{\text{traj}}^{L,T,\eta,z_0}$ follow from the corresponding properties of the ℓ^1 norm. For the triangle inequality, for any contexts C, D, E and each time t ,

$$\|\pi_t^C - \pi_t^E\|_1 \leq \|\pi_t^C - \pi_t^D\|_1 + \|\pi_t^D - \pi_t^E\|_1.$$

Multiplying by $\frac{1}{2}$ and averaging this inequality over $t = 0, \dots, T$ gives

$$D_{\text{traj}}^{L,T,\eta,z_0}(C, E) \leq D_{\text{traj}}^{L,T,\eta,z_0}(C, D) + D_{\text{traj}}^{L,T,\eta,z_0}(D, E).$$

Also $D_{\text{traj}}^{L,T,\eta,z_0}(C, C) = 0$. It is a pseudometric rather than a metric because distinct contexts can generate identical trajectories under a fixed update rule and initial condition.

The value of $D_{\text{traj}}^{L,T,\eta,z_0}$ depends on the update rule, horizon, intensity, and initial condition because the trajectory (π_t^C) does. It is not determined by S alone: in the benchmark below, $S(C_B, C_A) = S(C_A, C_S) = 2/4$, but Table 12 gives very different dynamic distances for these pairs. Thus the structural and dynamic comparisons encode different information. $\square$

For the benchmark run with L equal to the exponential-replicator update, $\eta = 0.45$, $p_0 = q_0 = 0.5$, and $T = 50$, we obtain Table 12.

	Business C_B	Classroom C_A	Sport team C_S
Business C_B	0.000	0.385	0.373
Classroom C_A	0.385	0.000	0.018
Sport team C_S	0.373	0.018	0.000

Table 12: Dynamic trajectory distance D_{traj} between contexts for the benchmark run, normalized as an averaged total-variation distance in $[0, 1]$.

The structural and dynamic distances are not the same object. Nevertheless, in this benchmark the classroom and sport contexts are dynamically close, even though their structural sensitivity is $S(C_A, C_S) = 2/4$. The reason is that both dynamics concentrate rapidly on (G, U) . This illustrates why the paper distinguishes class structure from strategic and population behavior: contexts may differ in static classification but generate similar long-run behavior under a given learning rule.

11.7 Reproducibility of numerical results

The repository separates the model (`src/context_games`), tests, experiments, notebook, and generated results. The notebook imports the tested package and is not a second implementation. The reproduction command first runs the unit and regression test suite, then regenerates all CSV tables and vector figures, and finally executes the source notebook to a separate artifact without modifying it. Continuous integration repeats the workflow under CPython 3.12 with a fully pinned environment.

The deterministic trajectories require no seed. Their horizon is checked at $T \in \{25, 50, 100, 200, 500\}$ using both the one-step residual and the change from the preceding horizon. At $T = 50$ the maximum one-step residual over the three baseline contexts is 6.2×10^{-11} , and the largest change from $T = 50$ to $T = 100$ is 1.7×10^{-10} . The payoff-perturbation audit is stochastic only as a numerical integration over an explicitly specified synthetic distribution and uses the fixed seed 20260622. The exhaustive theorem audit is deterministic. All benchmark inputs, code, tests, and generated results are deposited with the article; no external or confidential data are used.

12 Methodological protocol for empirical calibration

Although the benchmark payoffs above are synthetic, the framework suggests a clear empirical route. A calibration study can proceed through contextual vignettes. A minimal pilot design would contain the following elements.

- (1) **Profiles.** Select a family of abstract profiles, such as constructive guidance, public pressure, resistance, cooperative uptake, sanction, strategic sacrifice, and information withholding.
- (2) **Contextual instantiation.** Instantiate each profile in several contexts: organization, classroom, sport team, public institution, or family. Each vignette should identify the authority-side actor, the affected side, the action, and the response.
- (3) **Participants and estimands.** Define the target population, sampling frame, primary contrasts, exclusion rules, and smallest substantively relevant class-margin or context effect before recruitment. Determine sample size from a preregistered precision or power analysis that accounts for repeated judgments and clustering by participant and vignette; no universal pilot size is defensible.
- (4) **Measures.** Ask participants to estimate the gain or loss for the acting player and for the affected player on a scale from -5 to 5 . Additional items may measure perceived fairness, legitimacy, humiliation, learning, compliance, trust, or cohesion.
- (5) **Normalization.** Normalize responses into payoff vectors (x, y) and compute induced benefit–loss classes.
- (6) **Uncertainty.** Use a hierarchical model or a cluster-respecting bootstrap to quantify uncertainty in both payoff coordinates. Report joint regions and the probability or confidence with which a profile lies on each side of an axis; marginal intervals alone do not identify quadrant membership.
- (7) **Hypotheses.** Preregister tests of context-by-profile contrasts, including whether profiles such as (P, U) cross a specified axis and whether contextual sensitivity differs across domains. Correct for multiplicity or designate a single primary contrast.

Such a study would test whether the theoretical phenomenon illustrated in Table 6 appears in human judgments. It would also allow different communities to have different valuation maps without changing the abstract strategic form. In such a study, the relevant empirical question is not whether a person is destructive, but whether a community evaluates a profile as producing non-compensating harm.

13 Discussion

The model developed here clarifies a distinction that is easily missed. A strategic form describes who can do what. A contextual valuation describes what those actions mean in a given domain. The benefit–loss class is induced only after valuation. Therefore, class is neither purely strategic nor purely subjective. It is relational: it depends on the interaction between an abstract profile and a context-specific interpretation of gains and losses.

The partition-based formulation strengthens the original sign-plane intuition. Each context induces a partition of the profile space. Class-preserving translations are exactly those translations whose inverse images preserve the induced partition, making explicit what is retained when a strategic form is transported between contexts.

The pseudometric S provides a quantitative comparison of contexts at the class level. It ignores payoff magnitude and focuses on whether the induced relational class changes. This is appropriate when the question is not how much the payoffs change, but whether the social meaning of the interaction changes. The robustness theorem complements this by showing when class invariance is stable under perturbation: profiles far from the axes cannot be recontextualized by small valuation changes.

The game-theoretic layer adds a second distinction. An intelligent profile need not be a Nash equilibrium, and a malicious profile may be one. In the business benchmark, (P, U) is malicious but admits no profitable unilateral deviation. In the sport benchmark, (P, U) is intelligent but not an equilibrium. Thus class desirability and equilibrium membership should be studied jointly rather than conflated.

The population-dynamics illustration adds a third distinction. Static contextual sensitivity and dynamic trajectory distance need not agree. Two contexts may classify some profiles differently but converge to similar long-run behavior under a given update rule. Conversely, contexts with similar class maps may behave differently if the payoff magnitudes and ordinal comparisons induce different dynamics. This motivates later work connecting class-level metrics, equilibrium structure, and population dynamics more systematically.

14 Limitations and future work

The paper has several limitations. First, the numerical payoffs are constructed examples, not measurements of organizations, classrooms, or teams; robustness to perturbation does not confer external validity. Second, seven features encode four profiles, so the weight representation is saturated and non-identified. Third, the exponential-replicator illustration is deterministic, low-dimensional, and conditional on independent random matching. Its invariant faces make every pure vertex a rest point, including non-equilibria, and it contains neither mutation nor estimated behavioral noise. Fourth, the perturbation rates describe a chosen uniform distribution over synthetic payoffs and are computational diagnostics rather than sampling statements. Fifth, boundary cases require domain-specific conventions, especially when one player is neutral. Finally, several partition observations are elementary; the substantive contribution rests on connecting class robustness to equilibrium configuration rather than on those observations in isolation.

These limitations delimit the scope of the framework. Empirical calibration, probabilistic valuation, more parsimonious identified features, networked interaction, and alternative learning

rules are natural next steps. Until such evidence exists, the domain labels should be read only as interpretations of synthetic games.

15 Conclusion

This paper proposed context-dependent benefit–loss games as a formal framework for studying benefit–loss interactions across domains. The main idea is simple but consequential: the benefit–loss class of a strategic profile is not an intrinsic property of the profile alone. It is induced by a context-dependent valuation map. By distinguishing strategic form from contextual valuation, the framework explains how the same abstract action may remain invariant across contexts or be reclassified when moved from one domain to another.

The formal contributions include induced partitions of profile spaces, a characterization of class-preserving translations, a Hamming pseudometric for contextual sensitivity, and exact robustness radii for class and pure-equilibrium sets. The strategic contribution shows that class invariance and Nash equilibrium membership are distinct, and the 2×2 configuration theorem separates a forced knife-edge regime from a possible robust diagonal regime. The computational audits challenge these claims through exhaustive finite enumeration, seeded perturbations, and convergence diagnostics.

The benchmark examples show how these notions operate in three domains: business organization, classroom, and sport team. The central transition $(P, U) : \mathbf{M} \rightarrow \mathbf{D} \rightarrow \mathbf{I}$ demonstrates that contextual valuation can move a profile from malicious, to destructive, to intelligent while leaving the abstract strategic form unchanged. The second transition $(G, R) : \mathbf{D} \rightarrow \mathbf{H} \rightarrow \mathbf{D}$ shows that recontextualization is not always a movement toward improvement. This paper provides the structural and computational layer of that program: empirical calibration is not required for the validity of the formal results, but is the natural next step for domain-specific applications. Future work should estimate contextual valuations empirically, extend the framework to larger games and networks, and connect class transitions with dynamical models of trust, cohesion, performance, and accumulated harm.

Data and code availability

All synthetic benchmark inputs, source code, tests, and scripts needed to reproduce the tables and figures are available at <https://github.com/cero1979/GamesAndContext>. The study uses no confidential or personal data. A versioned archival DOI will be added before submission.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During revision, the author(s) used OpenAI Codex to assist with code refactoring, test construction, reproducibility checks, and language editing. The author(s) reviewed and edited all outputs, verified the mathematical and computational claims, and take full responsibility for the content of the article.

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